

Nexidia Data Exchange Framework

Installation Guide



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CONTENTS

1: Introduction	3
Document Revision History	4
2: Prerequisites	7
3: Installing the Application	9
NEAT/File System Installation	11
NEAT/SFTP Installation	19
NEST/File System Installation	21
NEST/SFTP System Installation	26
Compute Node Installation	30
Single Server Installation	35
Installing Multiple Instances of DEF on the Same Server	43
NEAT Server Installation	43
NEST Server Installation	57
Multi-Server Installation using Config File	64
Silent Installation	66
SilentInstaller.properties File	69
4: Migrating DEF 3.5 to DEF 5.x	73
Required DEF 3.5 Information	73
Installing DEF 5.x	74
Porting Processes from DEF 3.5 to DEF 5.x	75
5: Sentinel Monitoring	77
Returned Metrics	77
Sample JSON Output	78

6: Upgrading Data Exchange Framework	81
Validating Custom Actions	82
Upgrading an Individual Server	87
Upgrading Multiple Servers	94
Upgrading a Clustered DEF Environment	96
Stop the DEF Processes	97
Run the Cumulative Updater to Upgrade a Clustered DEF Environment	98
Restart the DEF Processes	103
Verify the DEF Version	104
 7: Uninstalling the Application	 105
 8: Troubleshooting	 111
Common Installation Issues	111
Diagnostic Tools	113

Introduction

Nexidia Data Exchange Framework manages file movement and data processing from customer locations to the Interaction Analytics secure environment, to provide simplified processing for extracting data from customer sites.

You can deploy DEF in either hosted or dedicated environments. It includes two server types - Administration (NEAT) servers, and Worker (NEST) servers. An typical deployment has one Administration server with many associated Worker servers. The Worker servers report the processing status to the Administration server via a heartbeat message.

DOCUMENT REVISION HISTORY

SFTWARE VERSION	DATE	DESCRIPTION
12.5.1	July 2025	
12.5	December 2024	No updates.
12.4.2	May 2024	No updates.
12.4.2	March 2024	No Updates.
12.4.1	July 2023	No updates.
12.4	April 2023	• Upgrading a Clustered DEF Environment on page 96: Add this new topic.
5.10 FP1	November 2022	• NEAT/File System Installation on page 11: Updated the Configure Server step to include new default values for the Server retention (days) and Audit log retention (days) settings.
5.10	August 2022	• Installing the Application on page 9: Added the new Single Server installation option for installing DEF as a single server • Single Server Installation on page 35: Added a new procedure for installing DEF as a single server. • SilentInstaller.properties File on page 69: Added a new supported ServerType value, Single Server, for installing DEF as a single server.

SOFTWARE VERSION	DATE	DESCRIPTION
5.9	May 2022	• NEAT/File System Installation on page 11: Updated the procedure to include a new option for disabling auto-tuning. • Upgrading Data Exchange Framework on page 81: Added a note that when you upgrade your DEF installation, the Administration (NEAT) server and all Worker (NEST) servers must be upgraded to the same DEF version.
5.8	April 2022	• Upgrading an Individual Server on page 87 and Upgrading Multiple Servers on page 94: Noted that when you upgrade the Data Exchange Framework, the Cumulative Updater automatically replaces the <i>log4j.properties</i> file with the <i>log4j2.properties</i> file. • Validating Custom Actions on page 82: Added this new topic explaining how you can prepare for an upgrade by validating that the classes in your custom actions are compatible with the version of Data Exchange Framework to which you are upgrading. • Upgrading an Individual Server on page 87: Removed references to the deprecated Preserve Existing Patch File option, and replaced them with details about the new Custom Action Jar option.

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Prerequisites

- Each DEF installation requires a minimum of 16GB of RAM.
- To prevent validation errors and other issues, verify that you have met all of the requirements in the *Nexidia Data Exchange Framework System Requirements Guide*.
- Verify that you have created the Administration (NEAT) database in the SQL Server instance you are using. The database you create must be owned by the DEF service account user.
- You must be logged in and running the installer with administrator permissions.
- Verify that the target install machine has enough RAM. See the *Nexidia Data Exchange Framework System Requirements Guide* for more information.
- The Data Exchange Framework and Interaction Analytics must be installed on separate machines.
- When running the Install.bat file, make sure you run it in a plain DOS window. Log in using the service account under which the DEF will run, or by running Install.bat as that user (using the Windows “run as” command).
- If the DEF needs to connect to a database (such as Interaction Analytics), make sure there is connectivity between the them, and that you have the correct database credentials.
- If the DEF needs access to shared network folders (such as Interaction Analytics ingest folders), make sure there is connectivity between them.
- You can install DEF to run using the default system account or a specified system account with permissions to access the DEF data directories.
- If you are upgrading your DEF installation, the Administration (NEAT) server and all Worker (NEST) servers must be upgraded to the same DEF version.

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Installing the Application

Each DEF installation must include an Administration (NEAT) server and at least one Worker (NEST) server. You can expand the system by adding additional NEST servers.

DEF has several installation options:

- **NEAT in File System mode:** The Administration server communicates with its servers through the file system (either local, or a CIFS share). In this mode, the folder information is entered into the sites, and the communication subdirectories are entered into the servers.
- **NEAT in SFTP mode:** The Administration servers communicate with its servers over SFTP. In this mode, the SFTP connection information is entered into the sites, and the communications subdirectories are entered into the servers.
- **NEST in File System mode:** The Worker server communicates with its administration server through a file system folder. You will enter the connection information during installation.
- **NEST in SFTP mode:** The Worker server communicates with its administration server over SFTP. You will enter the connection information during installation.
- **Compute Node mode:** Facilitates horizontal scaling of a Worker server in response to increased workload or backlog. This eliminates the overhead of configuring and balancing the load across Worker servers. During installation, you will point the compute node at the install and data directories of the Worker server.
- **Single Server mode:** The Administration server and a Worker server are installed on a single server. This environment minimizes the DEF footprint and provides minimal content retention. A separate database server should be used for this type of environment.

Tip

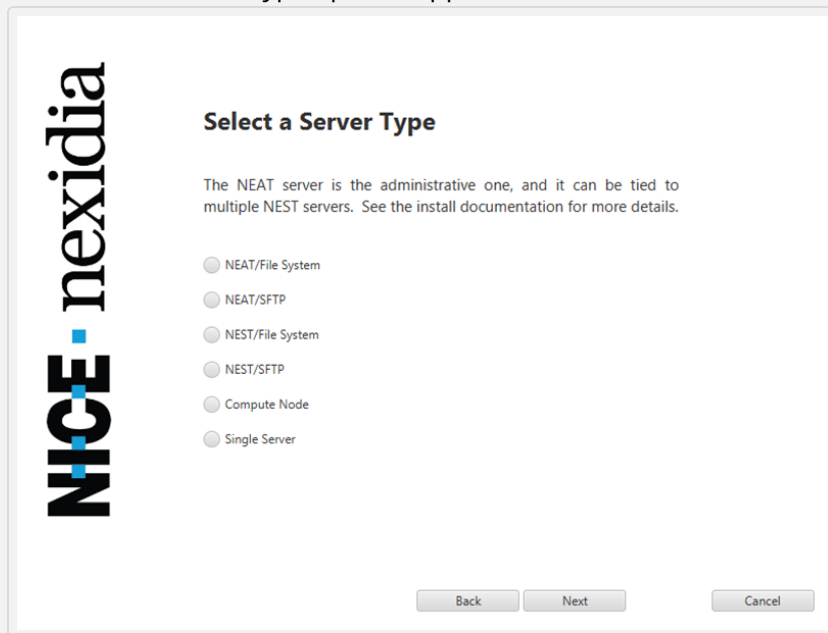
You can install as many NEAT server instances as required. The services will be named NDEFA1, NDEFA2, and so on. Ensure that the server size is large enough to support the number of NEAT server instances you intend to run.

NEAT/FILE SYSTEM INSTALLATION

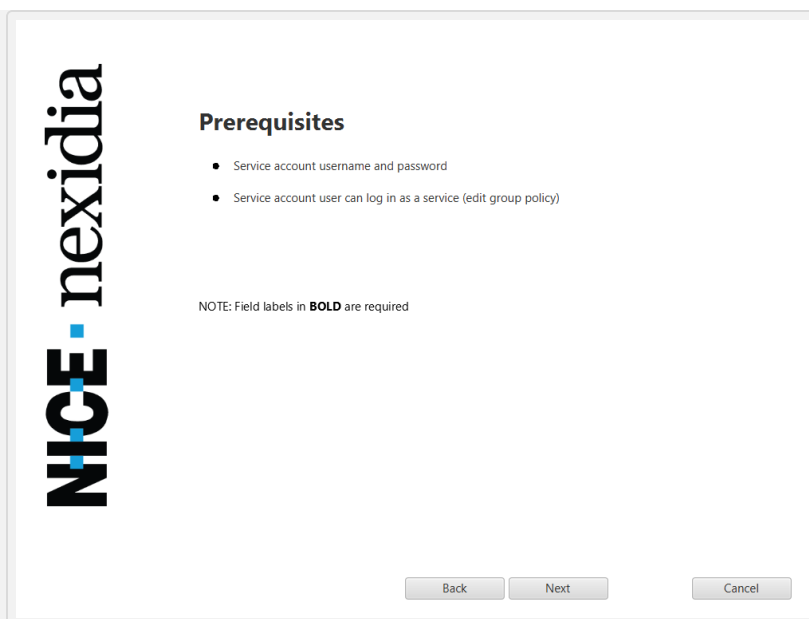
In NEAT/File System mode, the Administration server communicates with its servers through the file system (either local, or a CIFS share). In this mode, the folder information is entered into the sites, and the communication subdirectories are entered into the servers.

→ To install a NEAT File System:

1. Unzip the *DataExchangeFramework-Install-X.X.zip* file to the **root\.....\DataExchangeFramework-Install-X.X\Data Exchange Framework\install** directory.
2. Run the *Install.bat* file as an administrator.
The installation wizard launches, and prompts you to specify if you want to use the embedded JRE.
3. Enter **y** to use the embedded JRE, or **n** to use JAVA_HOME. If you don't want to use the embedded JRE and you have not set up your JAVA_HOME, enter the path to the JRE.
The Select a Server Type options appear.

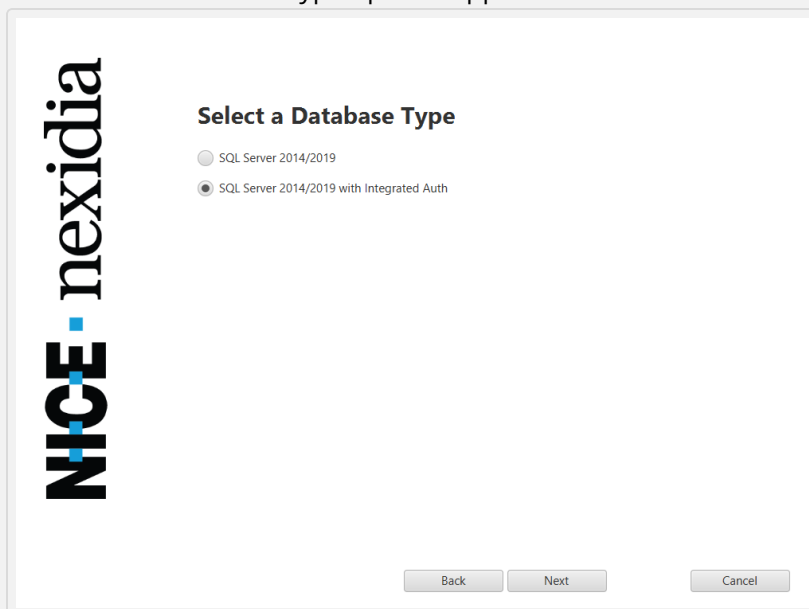


4. Select **NEAT/File System** and click **Next**.
The Prerequisites options appear.

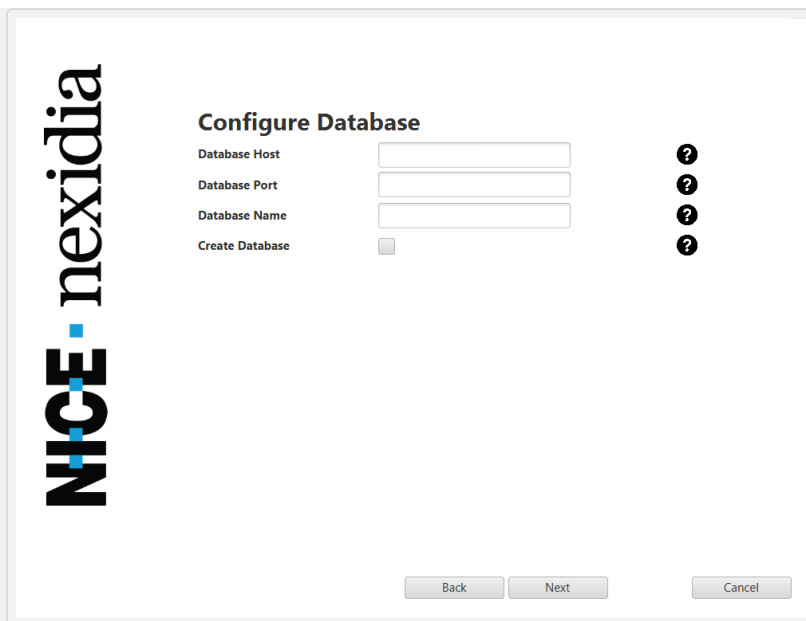


5. Click **Next**.

The Select a Database Type options appear.



6. Select the database server type, then click Next.
The Configure Database options appear.



NICE nexidia

Configure Database

Database Host ?

Database Port ?

Database Name ?

Create Database ☐ ?

Back Next Cancel

7. Enter the following information:
 - **Database Host:** The server on which the database resides.
 - **Database Port:** The port for accessing the database. The default is 1433 for the SQL server. If the SQL service is a named instance using a dynamic port, leave this field empty.
 - **Database Name:** The name of the database .
 - **Create Database:** Select this checkbox to create the database if it does not already exist.
8. Click **Next**.
The Configure Service options appear.

NICE nexidia

Configure Service

Service Account User

Service Account Password

☐ Show Password

Password Confirmation

Path to Data Folder

9. Enter the following information:

- **Service Account User:** The Service Account username.
- **Service Account Password:** The Service Account password.
- **Path to Data Folder:** The path to the data folder in which the Workbench files and other DEF-related data files are stored.

10. Click **Next**.

The Configure Server options appear.

Configure Server - NEAT/File

HTTP Port	8080	?
HTTPS Port	8181	?
Shutdown Port	8078	?
Default Threads Per Action	4	?
Server retention (days)	45	?
DEF Internal log retention (days)	14	?
Audit log retention (days)	14	?
Disable Thread Count Auto-Tuning	☐	?

Back Next Cancel

11. Enter the following information:

- **HTTP Port:** For connection to the NEAT user interface via the browser. The default value is **8080**.
- **HTTPS Port:** For connection to the NEAT user interface via the browser. The default value is **8181**.
- **Shutdown Port:** For shutdown requests from the Windows service. The default value is **8078**.
- **Default Threads Per Action:** The default server-level threads per action value. It is best to set to 70% of the available processors on the NEST servers. The default value is **4**.
- **Server retention (days):** The number of days beyond which it's safe to delete the DEF internal database, contents of auto-provisioned file system resources, old rows in all working tables' rollover, and quarantine tables. The default value is **45**.

Automated cleanup of these items takes place nightly. Active working table cleanup is included in the nightly run, but it uses the maximum self-reconciliation days setting at the process level, and ages out from there into the corresponding `{working table}$Q` quarantine table. This setting then determines the aging out from there.

- **DEF Internal log retention (days):** The number of days beyond which it's safe to delete the DEF internal logs. The default value is **14**. Automated cleanup of the internal logs takes place nightly.
- **Audit log retention (days):** The number of days beyond which it's safe to delete old audit log files. The default value is **14**. Automated cleanup of the audit logs takes place nightly. Audit logs are distinct from regular logging, and only include output from those actions that require it, like the NMF Decrypt suite, or the File Mover when it's in auditing mode for specific extensions.
- **Disable Thread Count Auto-Tuning:** Select this checkbox to disable thread count auto-tuning so that all actions inherit the server-level thread count for all runs, and will need to be manually tuned for optimal performance.

Important

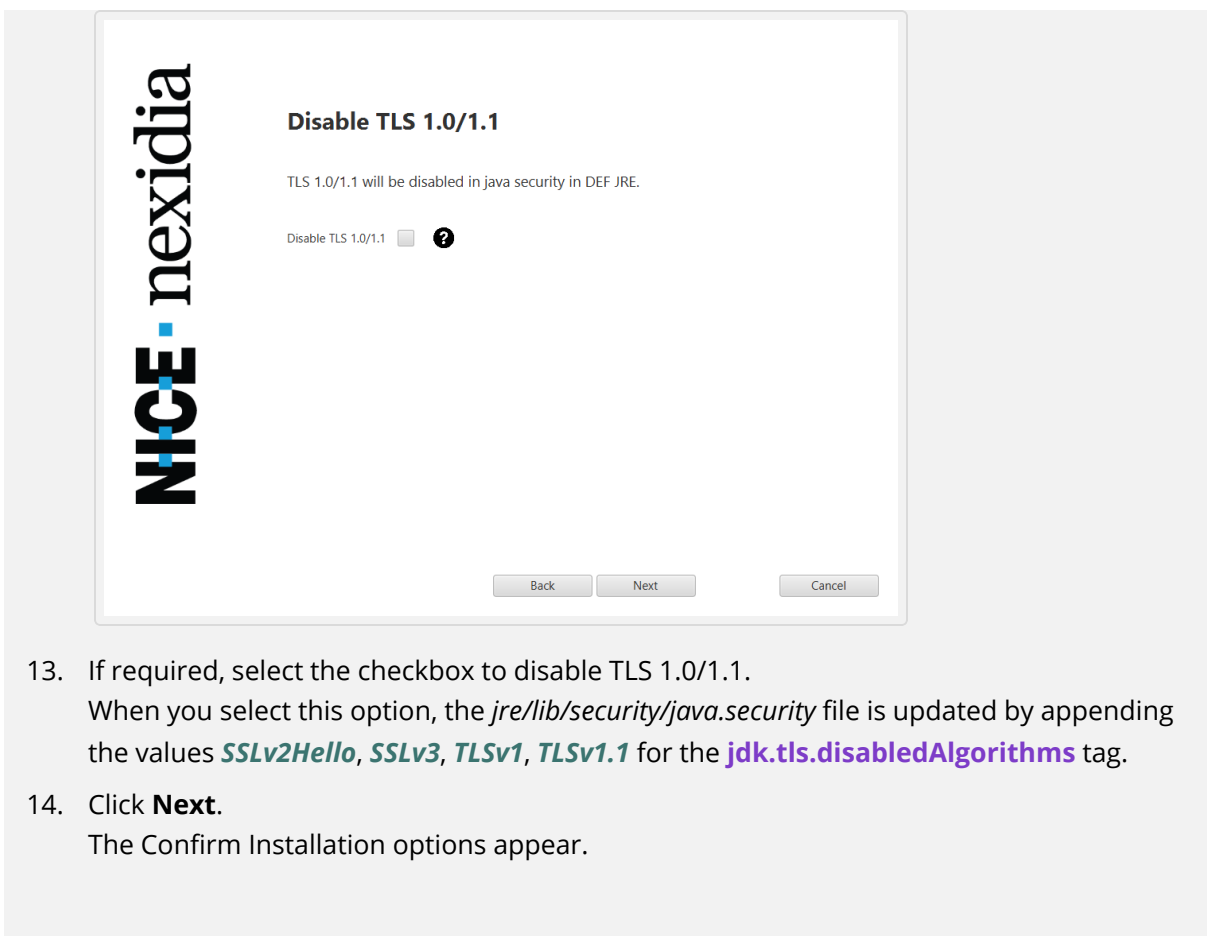
This option is not selected by default, and is only recommended for DEF installations on proxy servers or other servers where DEF contends heavily against other applications for system resources. If you select this option, a warning message appears to remind you that by selecting the option, you take responsibility for manually tuning the DEF actions.

12. Click **Next**.

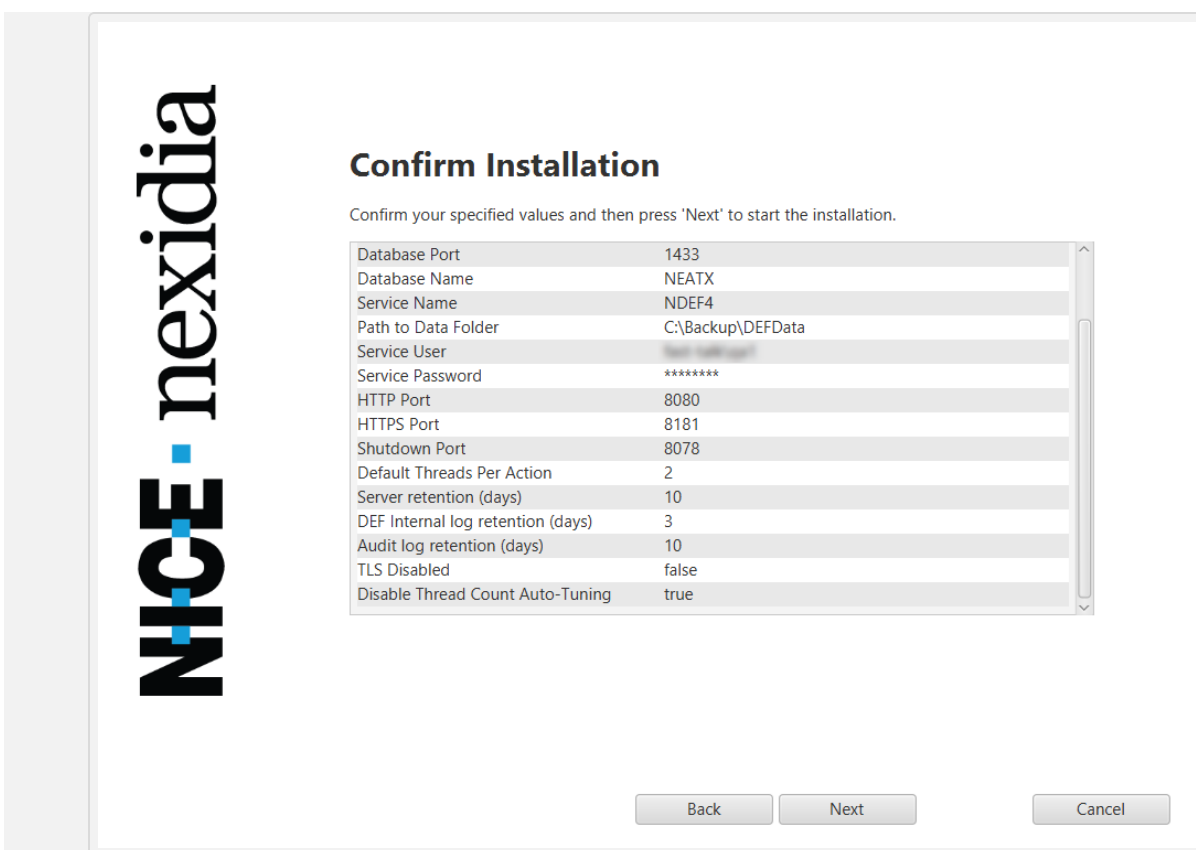
The Disable TLS 1.0/1.1 options appear.

Important

This option only appears if the embedded JRE folder doesn't already have TLS disabled for SSLv2Hello, SSLv3, TLSv1, TLSv1.1.



13. If required, select the checkbox to disable TLS 1.0/1.1.
When you select this option, the *jre/lib/security/java.security* file is updated by appending the values *SSLv2Hello*, *SSLv3*, *TLSv1*, *TLSv1.1* for the *jdk.tls.disabledAlgorithms* tag.
14. Click **Next**.
The Confirm Installation options appear.



15. Confirm the values and click **Next** to start the installation.
The Installation Status appears.
16. When the installation is complete, click **Next**.
The Installation Completed message appears. The database you created is added to the server Database folder, and the NDEFA1 service is added in the Services window.

Important

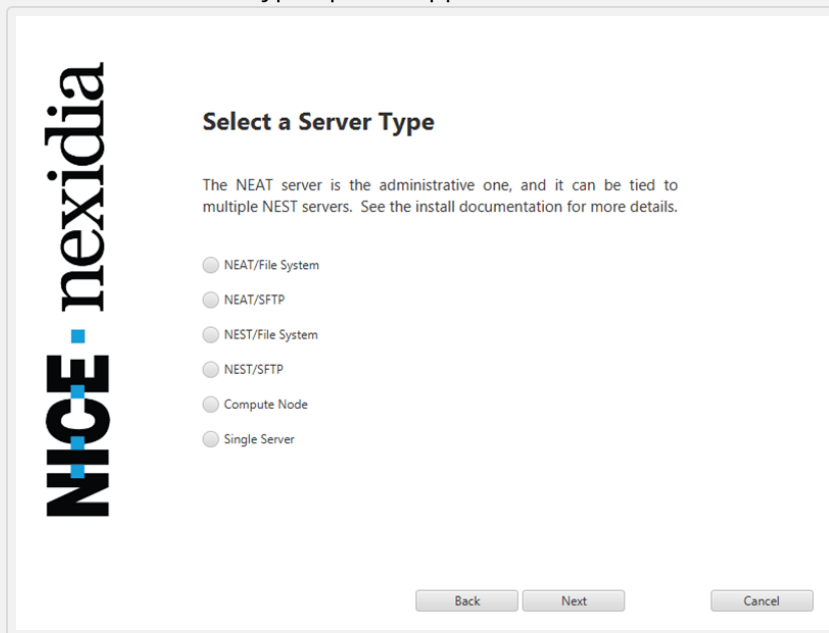
The DEF Administrator service is automatically named NDEFA1. This name cannot be changed. There can be as many NEAT instances as required. They will be named NDEFA1, NDEFA2, and so on.

NEAT/SFTP INSTALLATION

In NEAT/SFTP mode, the Administration server communicates with its servers over SFTP. In this mode, the SFTP connection information is entered into the sites, and the communications sub-directories are entered into the servers.

→ To install a NEAT/SFTP Installation:

1. Unzip the *DataExchangeFramework-Install-X.X.zip* file to the **root\.....\DataExchangeFramework-Install-X.X\Data Exchange Framework\install** directory.
2. Run the *Install.bat* file as an administrator.
The installation wizard launches, and prompts you to specify if you want to use the embedded JRE.
3. Enter **y** to use the embedded JRE, or **n** to use JAVA_HOME. If you don't want to use the embedded JRE and you have not set up your JAVA_HOME, enter the path to the JRE.
The Select a Server Type options appear.



4. Select **NEAT/SFTP** and click **Next**.

The Prerequisites options appear.

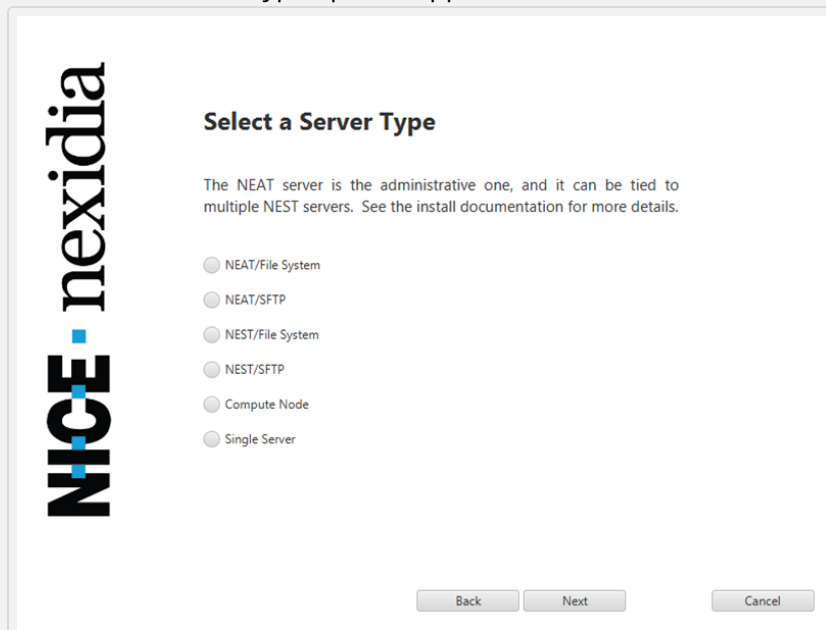
5. Complete the installation from step 5 in [NEAT/File System Installation](#) on page 11

NEST/FILE SYSTEM INSTALLATION

The Worker server communicates with its administration server through a file system folder. In this mode folder, information is entered during installation.

→ To install a NEST/File system:

1. Unzip the *DataExchangeFramework-Install-X.X.zip* file to the **root\.....\DataExchangeFramework-Install-X.X\Data Exchange Framework\install** directory.
2. Run the *Install.bat* file as an administrator.
The installation wizard launches, and prompts you to specify if you want to use the embedded JRE.
3. Enter **y** to use the embedded JRE, or **n** to use JAVA_HOME. If you don't want to use the embedded JRE and you have not set up your JAVA_HOME, enter the path to the JRE.
The Select a Server Type options appear.



4. Select **NEST/File System** and click **Next**.
The Prerequisites options appear.
5. Click **Next**.

The Configure Database options appear.

6. Enter the following information:

- **Database Host:** The server on which the database resides.
- **Database Port:** The port for accessing the database. The default is 1433 for the SQL server. If the SQL service is a named instance using a dynamic port, leave this field empty.
- **Database Name:** The name of the database .
- **Create Database:** Select this checkbox to create the database if it does not already exist.

7. Click **Next**.

The Configure Service options appear.

Configure Service - NEST/File

Service Account User: Service account username ?

Service Account Password: Service account password ?

☐ Show Password ?

Password Confirmation: Service account password ?

Path to Data Folder: Path to DEF data folder Browse ?

Back Next Cancel

8. Enter the following information:

- **Service Account User:** The Service Account username.
- **Service Account Password:** The Service Account password.

- **Path to Data Folder:** The path to the data folder in which the Workbench files and other DEF-related data files are stored.

9. Click **Next**.

The Configure Server options appear.

Configure Server - NEST/File

NEST Server ID: 117

File Comms Path: C:\DEFComms\ATL [Browse](#)

File Path Username (CIFS): File path username

File Path Password (CIFS): File path password

Shutdown Port: 8078

NOTE: CIFS username and password are optional

Back Next Cancel

10. Enter the following information:

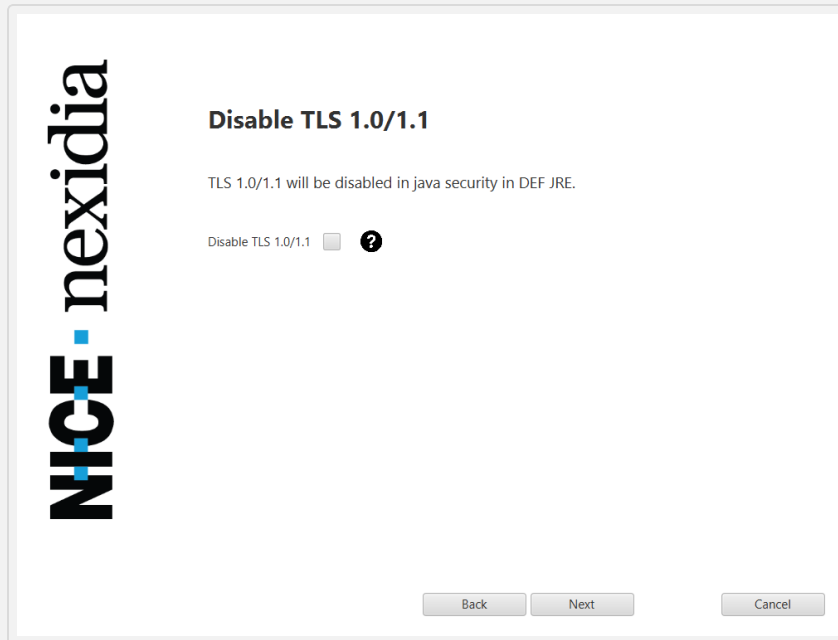
- **NEST Server ID:** From the NEAT server configuration.
- **File Path:** The path over which NEAT/NEST communications will happen. For NEST servers, this should be a concatenation of the site and server communication paths assigned in the NEAT user interface.
- (Optional) **File Path Username** (CIFS only)
- (Optional) **File Path Password** (CIFS Only)
- **Shutdown Port:** The port the Windows service uses to communicate shutdown requests to the DEF.

11. Click **Next**.

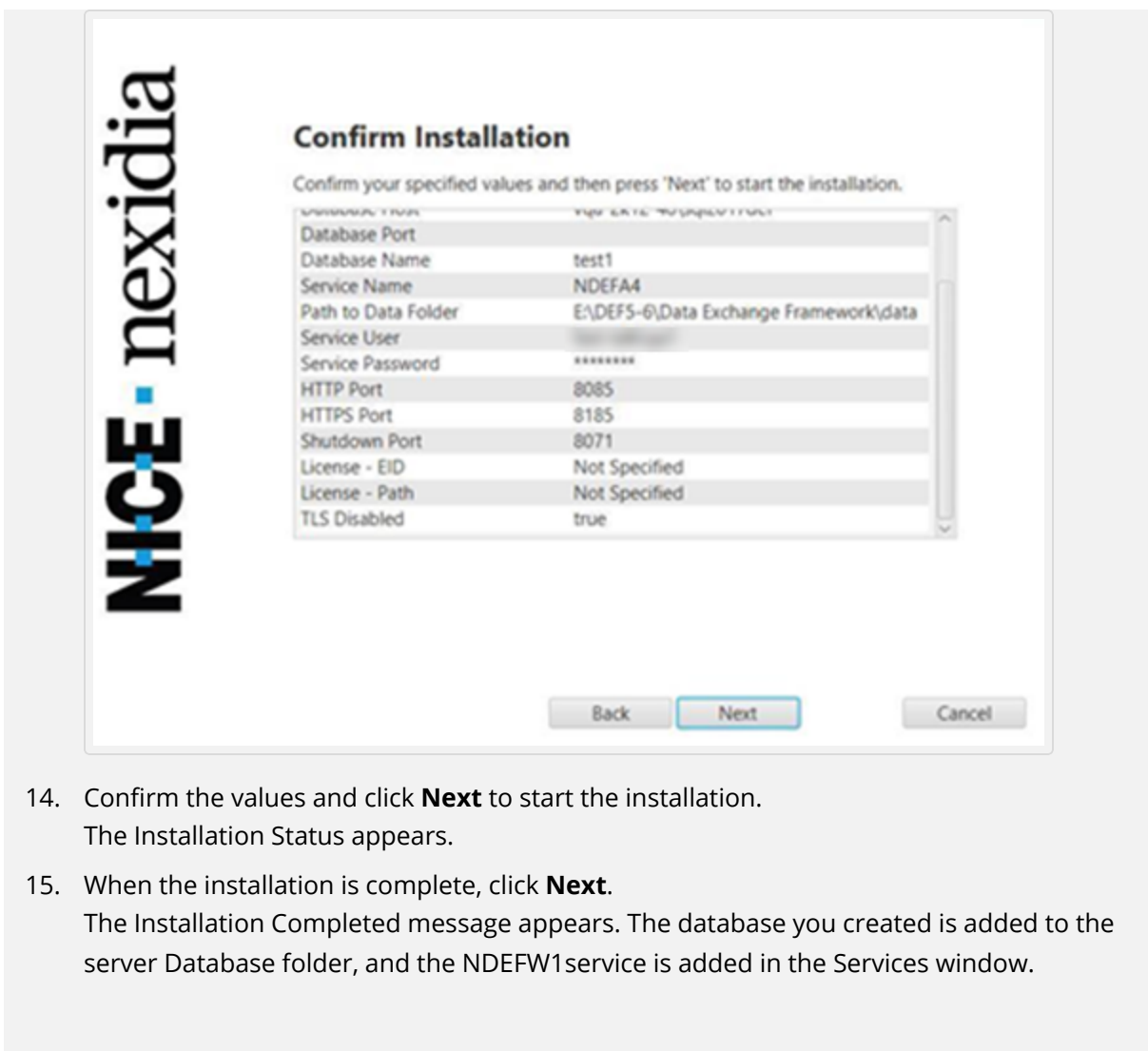
The Disable TLS 1.0/1.1 options appear.

Important

This option only appears if the embedded JRE folder doesn't already have TLS disabled for SSLv2Hello, SSLv3, TLSv1, TLSv1.1.



12. If required, select the checkbox to disable TLS 1.0/1.1.
When you select this option, the *jre/lib/security/java.security* file is updated by appending the values **SSLv2Hello**, **SSLv3**, **TLSv1**, **TLSv1.1** for the **jdk.tls.disabledAlgorithms** tag.
13. Click **Next**.
The Confirm Installation options appear.



14. Confirm the values and click **Next** to start the installation.
The Installation Status appears.
15. When the installation is complete, click **Next**.
The Installation Completed message appears. The database you created is added to the server Database folder, and the NDEFW1 service is added in the Services window.

Important

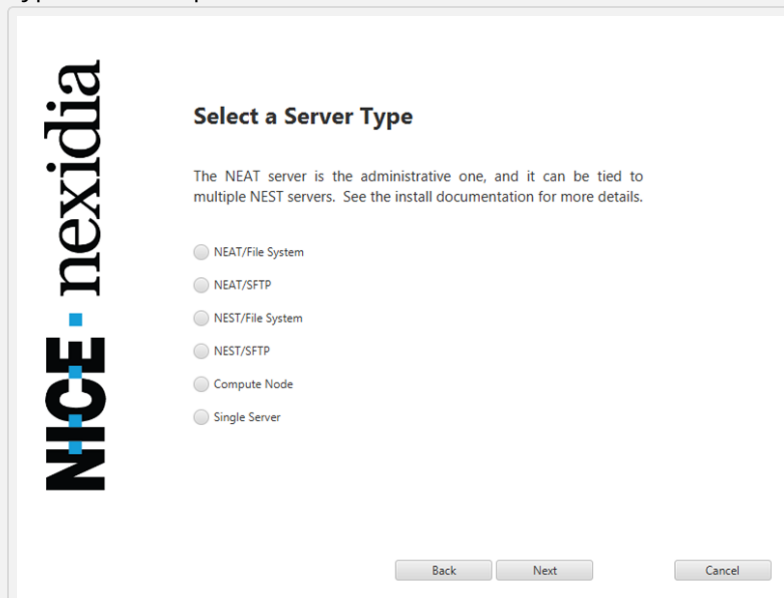
The DEF Worker service is automatically named NDEFW1. This name cannot be changed. There can be as many NEST instances as required. They will be named NDEFW1, NDEFW2, and so on.

NEST/SFTP SYSTEM INSTALLATION

The Worker server communicates with its administration server over SFTP. In this mode, the SFTP connection information is entered during installation.

→ To install a NEST/SFTP system:

1. Unzip the *DataExchangeFramework-Install-X.X.zip* file to the **root\.....\DataExchangeFramework-Install-X.X\Data Exchange Framework\install** directory.
2. Run the *Install.bat* file as an administrator.
The installation wizard launches, and prompts you to specify if you want to use the embedded JRE.
3. Enter “y” or “n”. If you don't want to use the embedded JRE, then you must have JAVA_HOME in your environment. If you have not set up your JAVA_HOME, you are asked to enter the path to the JRE. After you make the embedded JRE selection, the Select a Server Type window opens.



4. Select **NEST/SFTP** and click **Next**. A prerequisites window opens with the installation requirements. The requirements are also available in the Nexidia Data Exchange Framework System Requirements Guide.

5. Click **Next**.

The Configure Database window opens.

6. Continue to configure the data as described in [NEST/File System Installation](#) on page 21 until you reach the Configure Server - NEST/SFTP window shown below.

Configure Server - NEST/SFTP

NEST Server ID	1	?
SFTP Host		?
SFTP Port	22	?
SFTP Comms Path	/test	?
SFTP Username	developer	?
Shutdown Port	8078	?
SFTP Password Type	Password Authentication	?
SFTP Password	*****	?

Back Next Cancel

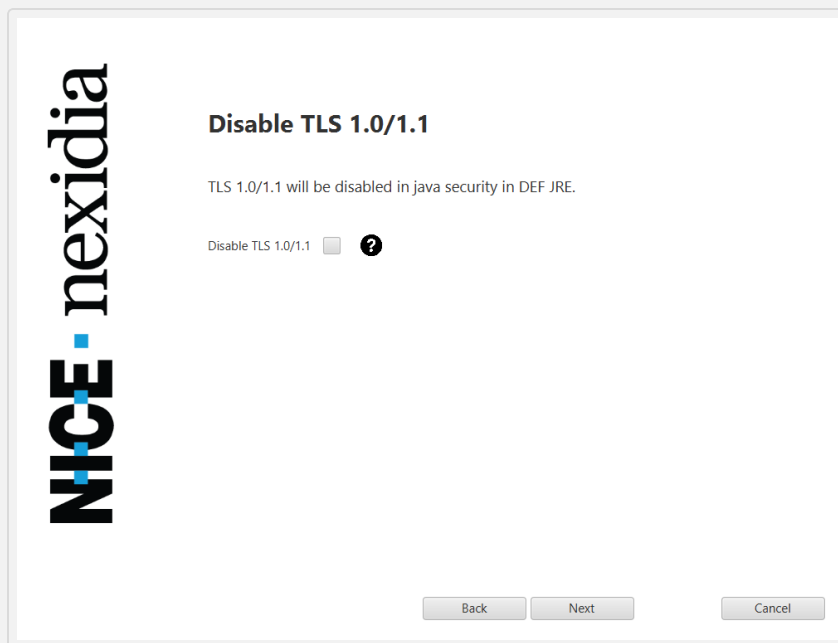
7. Enter information as follows:
 - **NEST Server ID** from the NEST server configuration
 - **SFTP Host** for NEAT/NEST communications
 - **SFTP Port**.
 - **SFTP Comms Path**. This is the path relative to logged in the users home directory for NEAT/NESTcommunications
 - **SFTP User Name**
 - **Shutdown Port**: for the Windows service to use to communicate shutdown requests to the DEF
 - **SFTP Password Type**:

- Password Authentication. If you select this option, enter the password authentication.
- Key-based Authentication. If you select this option, enter the SFTP Key Location and the SFTP Key Passphrase.

8. Click **Next**. The Disable TLS 1.0/1.1 window opens.

Important

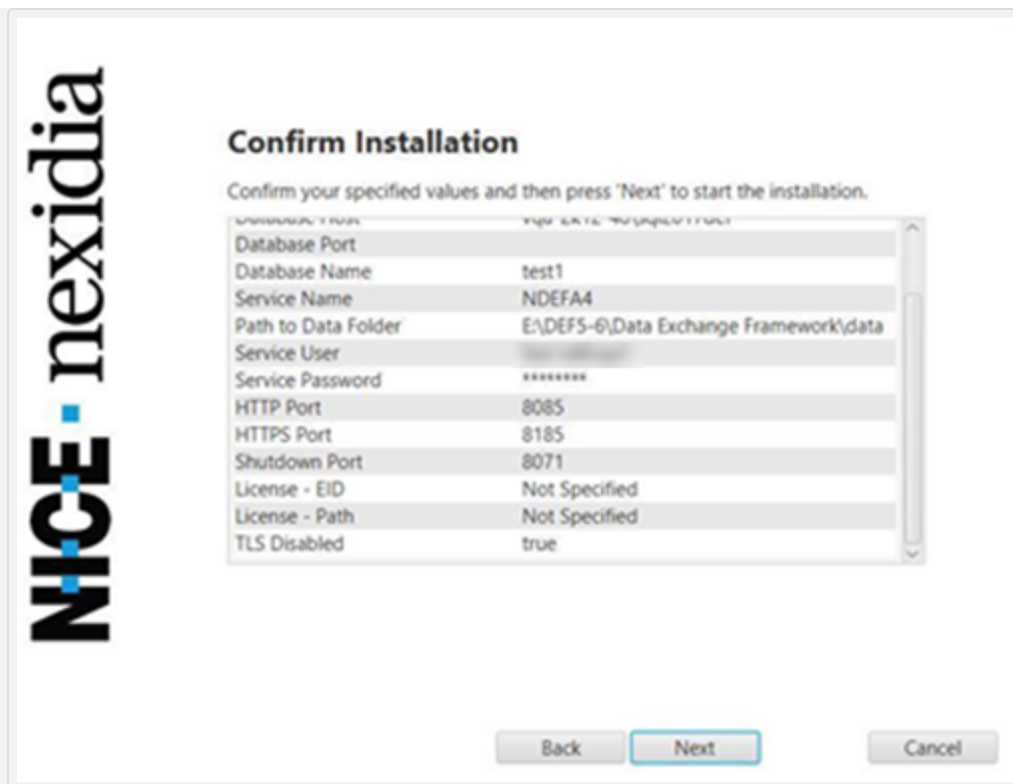
If External JRE is selected on the command prompt at the start of the install, then this window does not display. The window displays if embedded JRE folder doesn't already have the TLS disabled for SSLv2Hello, SSLv3, TLSv1, TLSv1.1. If these are already included in the embedded jre folder, then the TLS screen does not display.



9. Select the checkbox to disable TLS 1.0/1.1.

If Disable TLS is selected, it updates the `jre/lib/security/java.security` file by appending the values: `SSLv2Hello, SSLv3, TLSv1, TLSv1.1` for the tag: `jdk.tls.disabledAlgorithms`

10. Click **Next**. The Confirm Installation window opens.



11. Confirm the values and click **Next** to start the installation. The Installation Status window opens.
12. Click **Next**. The Installation Completed window opens. The database you created is added to server Database folder and NDEFW1 service is added in the Services window.

Important

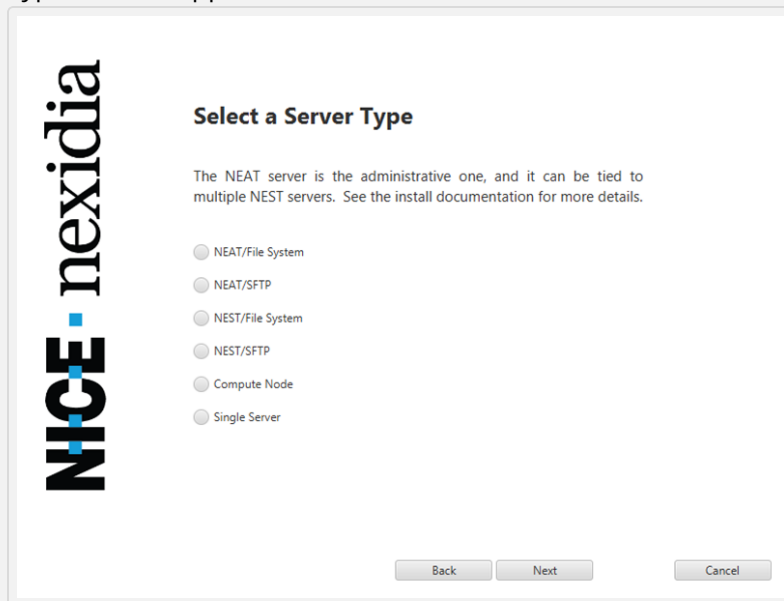
Nexidia Data Exchange Framework Worker1 is the name for the Worker service for DEF. This name is provided automatically and cannot be changed. There can be as many NEST instances as required. They will be named Nexidia Data Exchange Framework Worker1, Nexidia Data Exchange Framework Worker2 etc.

COMPUTE NODE INSTALLATION

Compute Node provides the ability to horizontally scale a Worker server quickly and easily in response to increased workload or backlog, without the overhead of configuring and balancing load across multiple Worker servers.

→ To install a compute node:

1. Unzip the *DataExchangeFramework-Install-X.X.zip* file to the **root\.....\DataExchangeFramework-Install-X.X\Data Exchange Framework\install** directory.
2. Run the *Install.bat* file as an administrator.
The installation wizard launches, and prompts you to specify if you want to use the embedded JRE.
3. Enter “y” or “n”. If you don't want to use the embedded JRE, then you must have JAVA_HOME in your environment. If you have not set up your JAVA_HOME, you are asked to enter the path to the JRE. After you make the emedded JRE selection, the Select a Server Type window appears.



4. Select **Compute Node** and click **Next**. A prerequisites window opens with the installation requirements.

5. Click **Next**. The Configure Service - Compute Node window opens.

Configure Service - Compute Node

Service Account User

Service Account Password

☐ Show Password

Password Confirmation

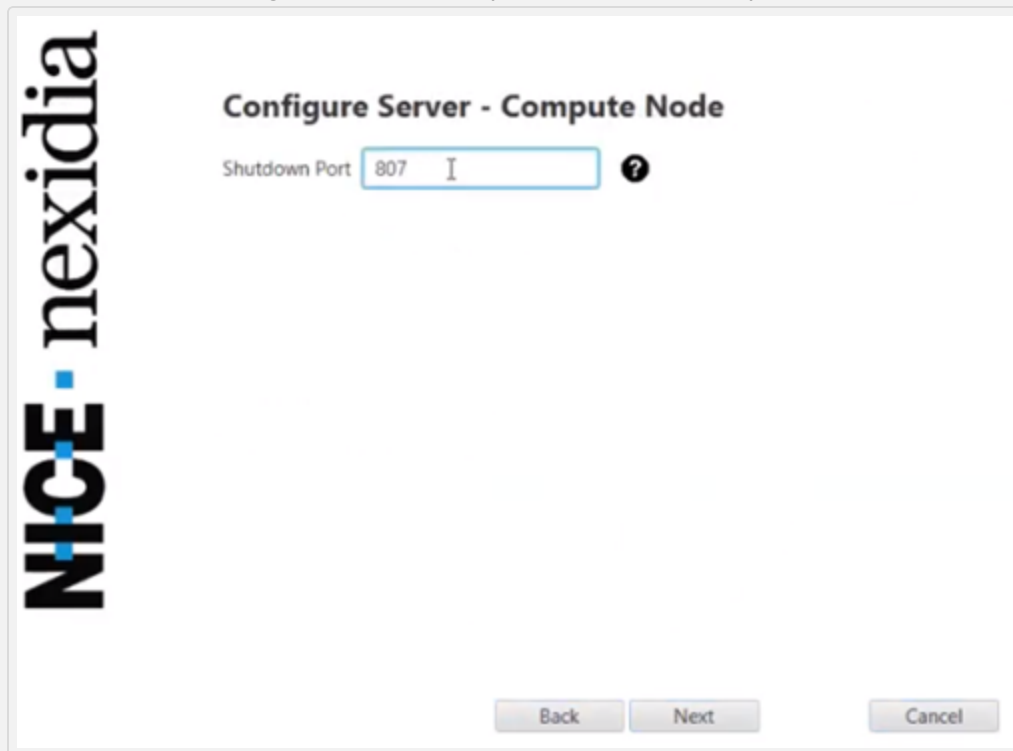
NEST Install Folder

NEST Data Folder

CN Local Data Folder

6. Enter information as follows:
- **Service Account User:** Windows account that the DEF will run under. It can be either a local user or a domain user in domain\username format
 - **Service Account Password and Confirmation:**
 - **NEST Install Folder:** Location of the compute node parent installation folder and must be a UNC path
 - **NEST Data Folder:** Location of the compute node parent data folder and must be a UNC path
 - **CN Local Data Folder:** This folder contain the compute node logs, audit logs, and Workbench sub folders
7. Click **Next**. The Compute Node - Select Processes Reminder window opens.

8. Click **Next**. The Configure Server - Compute Node window opens.



9. Enter the **port** the Windows service uses to communicate shutdown requests.
10. Click **Next**. The Disable TLS 1.0/1.1 window opens.

Important

If External JRE is selected on the command prompt at the start of the install, then this window does not display. The window displays if embedded JRE folder doesn't already have the TLS disabled for SSLv2Hello, SSLv3, TLSv1, TLSv1.1. If these are already included in the embedded jre folder, then the TLS screen does not display.



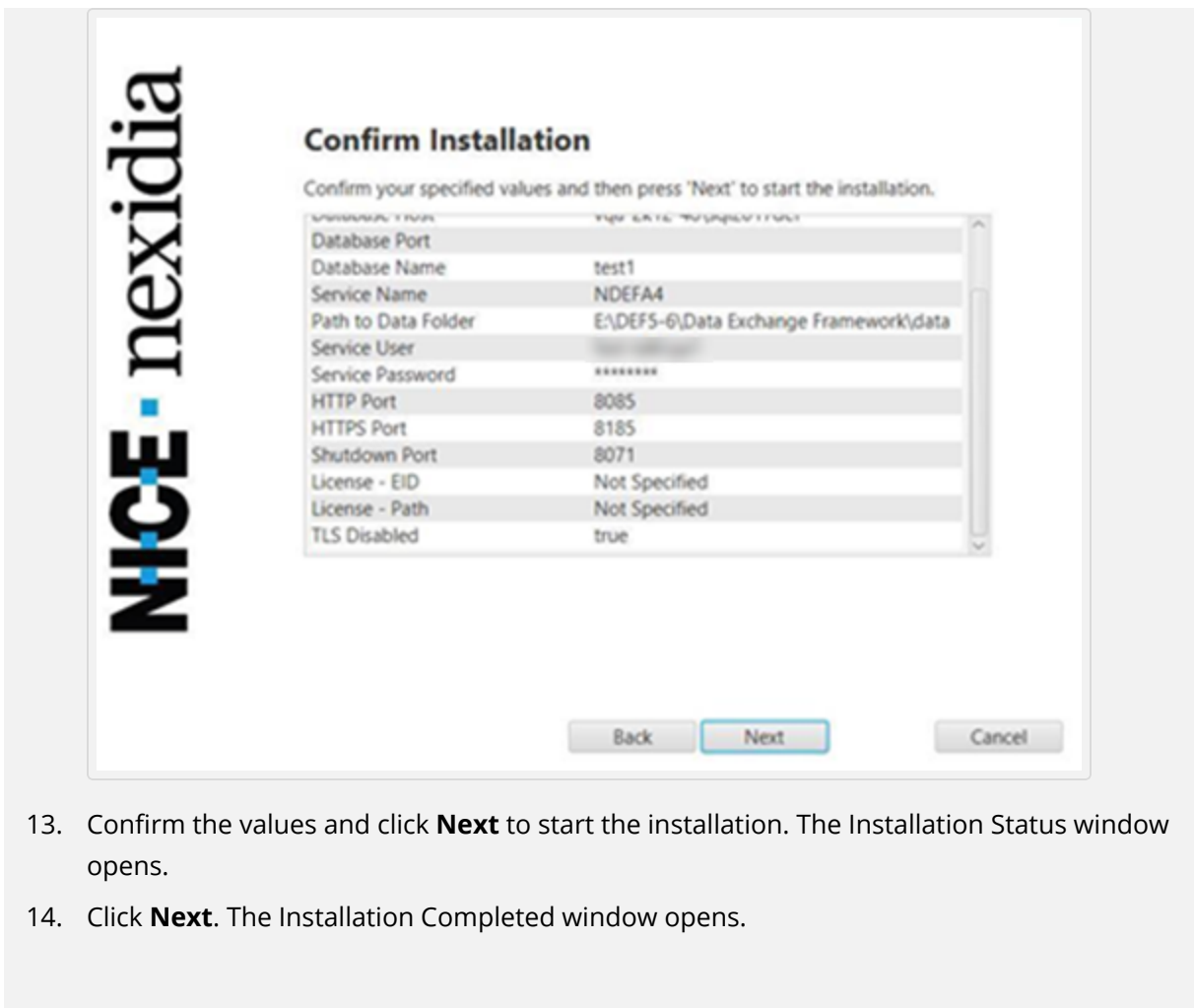
Disable TLS 1.0/1.1

TLS 1.0/1.1 will be disabled in java security in DEF JRE.

Disable TLS 1.0/1.1 ☐ ?

Back Next Cancel

11. Select the checkbox to disable TLS 1.0/1.1.
If Disable TLS is selected, it updates the `jre/lib/security/java.security` file by appending the values: `SSLv2Hello`, `SSLv3`, `TLSv1`, `TLSv1.1` for the tag: `jdk.tls.disabledAlgorithms`
12. Click **Next**. The Confirm Installation window opens.



13. Confirm the values and click **Next** to start the installation. The Installation Status window opens.
14. Click **Next**. The Installation Completed window opens.

Important

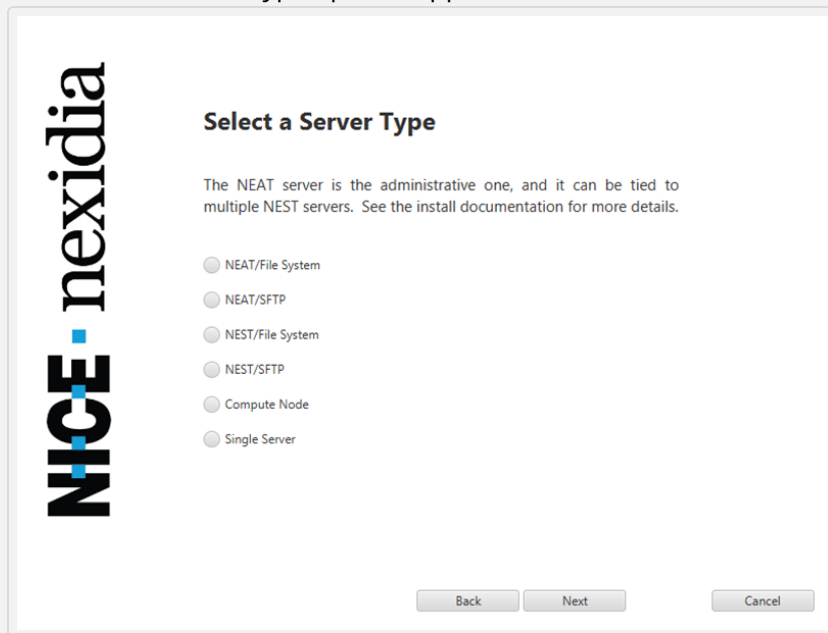
Nexidia Data Exchange Framework Compute Node1 is the name for the Worker service for DEF. This name is provided automatically and cannot be changed. There can be as many NEST instances as required. They will be named Nexidia Data Exchange Framework Compute Node1, Nexidia Data Exchange Framework Compute Node2, etc.

SINGLE SERVER INSTALLATION

In Single Server mode, both an Administration (NEAT) and a Worker (NEST) server are installed on a single machine. This mode minimizes the DEF footprint and provides minimal content retention. Database access remains as usual, and it is expected that a separate database server will be used.

→ To install a single server:

1. Unzip the *DataExchangeFramework-Install-X.X.zip* file to the **root\.....\DataExchangeFramework-Install-X.X\Data Exchange Framework\install** directory.
2. Run the *Install.bat* file as an administrator.
The installation wizard launches, and prompts you to specify if you want to use the embedded JRE.
3. Enter **y** to use the embedded JRE, or **n** to use JAVA_HOME. If you don't want to use the embedded JRE and you have not set up your JAVA_HOME, enter the path to the JRE.
The Select a Server Type options appear.



4. Select **Single System** and click **Next**.

The Prerequisites options appear.



Prerequisites

- Service account username and password
- Service account user can log in as a service (edit group policy)

NOTE: Field labels in **BOLD** are required

Back Next Cancel

5. Click **Next**.

The Select a Database Type options appear.



Select a Database Type

☐ SQL Server 2014/2019

☒ SQL Server 2014/2019 with Integrated Auth

Back Next Cancel

6. Select the database server type, then click **Next**.

The Configure Database options appear.

NICE nexidia

Configure Database

Database Host

Database Port

Database Name

Create Database ☐

Back Next Cancel

7. Enter the following information:

- **Database Host:** The server on which the database resides.
- **Database Port:** The port for accessing the database. The default is 1433 for the SQL server. If the SQL service is a named instance using a dynamic port, leave this field empty.
- **Database Name:** The name of the database .
- **Create Database:** Select this checkbox to create the database if it does not already exist.

8. Click **Next**.

The Configure Service - Single Server options appear.

NICE nexidia

Configure Service - Single Server

Service Account User

Service Account Password

☐ Show Password

Password Confirmation

Path to Data Folder

9. Enter the following information:

- **Service Account User:** The Service Account username.
- **Service Account Password:** The Service Account password.
- **Path to Data Folder:** The path to the data folder in which the Workbench files and other DEF-related data files are stored.

10. Click **Next**.

The Configure Server - Single Server options appear.

Configure Server - Single Server

HTTP Port	8080	?
HTTPS Port	8181	?
Shutdown Port	8078	?
Default Threads Per Action	2	?
Server retention (days)	10	?
DEF Internal log retention (days)	3	?
Audit log retention (days)	10	?
Disable Thread Count Auto-Tuning	Disabled	

Back Next Cancel

11. Enter the following information:

- **HTTP Port:** For connection to the NEAT user interface via the browser. The default value is 8080.
- **HTTPS Port:** For connection to the NEAT user interface via the browser. The default value is 8181.
- **Shutdown Port:** For shutdown requests from the Windows service. The default value is 8078.
- **Default Threads Per Action:** The default server-level threads per action value. The default value is 2. It is best to set to 70% of the available processors.
- **Server retention (days):** The number of days beyond which it's safe to delete the DEF internal database, contents of auto-provisioned file system resources, old rows in all working tables' rollover, and quarantine tables. The default value is 10.

Automated cleanup of these items takes place nightly. Active working table cleanup is included in the nightly run, but it uses the maximum self-reconciliation days setting at the process level, and ages out from there into the corresponding *{working table}* quarantine table. This setting then determines the aging out from there.

- **DEF Internal log retention (days):** The number of days beyond which it's safe to delete the DEF internal logs. The default value is 3. Automated cleanup of the internal logs takes place nightly.
- **Audit log retention (days):** The number of days beyond which it's safe to delete old audit log files. The default value is 10. Automated cleanup of the audit logs takes place nightly. Audit logs are distinct from regular logging, and only include output from those actions that require it, like the NMF Decrypt suite, or the File Mover when it's in auditing mode for specific extensions.

Important

Disable Thread Count Auto-Tuning is selected, and is read-only. This disables thread count auto-tuning, so that all actions inherit the server-level thread count for all runs.

12. Click **Next**.

The Disable TLS 1.0/1.1 options appear.

Important

This option only appears if the embedded JRE folder doesn't already have TLS disabled for SSLv2Hello, SSLv3, TLSv1, TLSv1.1.



13. If required, select the checkbox to disable TLS 1.0/1.1.
When you select this option, the *jre/lib/security/java.security* file is updated by appending the values **SSLv2Hello**, **SSLv3**, **TLSv1**, **TLSv1.1** for the **jdk.tls.disabledAlgorithms** tag.
14. Click **Next**.
The Confirm Installation options appear.



Confirm Installation

Confirm your specified values and then press 'Next' to start the installation.

Database Port	1433
Database Name	NEATX
Service Name	NDEF4
Path to Data Folder	C:\Backup\DEFData
Service User	*****
Service Password	*****
HTTP Port	8080
HTTPS Port	8181
Shutdown Port	8078
Default Threads Per Action	2
Server retention (days)	10
DEF Internal log retention (days)	3
Audit log retention (days)	10
TLS Disabled	false
Disable Thread Count Auto-Tuning	true

Back Next Cancel

15. Confirm the values and click **Next** to start the installation.
The Installation Status appears.
16. When the installation is complete, click **Next**.
The Installation Completed message appears.

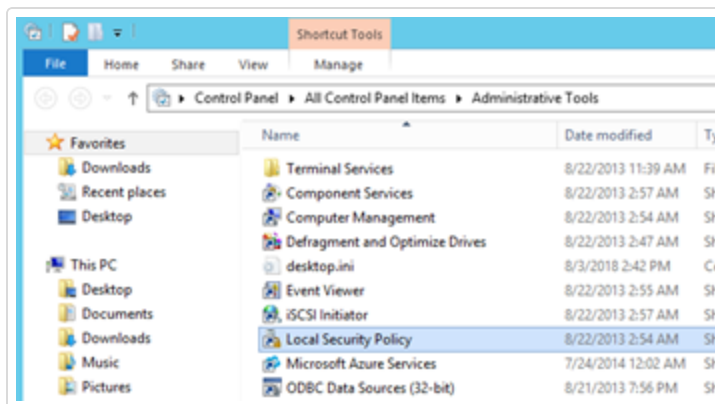
INSTALLING MULTIPLE INSTANCES OF DEF ON THE SAME SERVER

In this mode, multiple instances of DEF are installed on the same server.

NEAT Server Installation

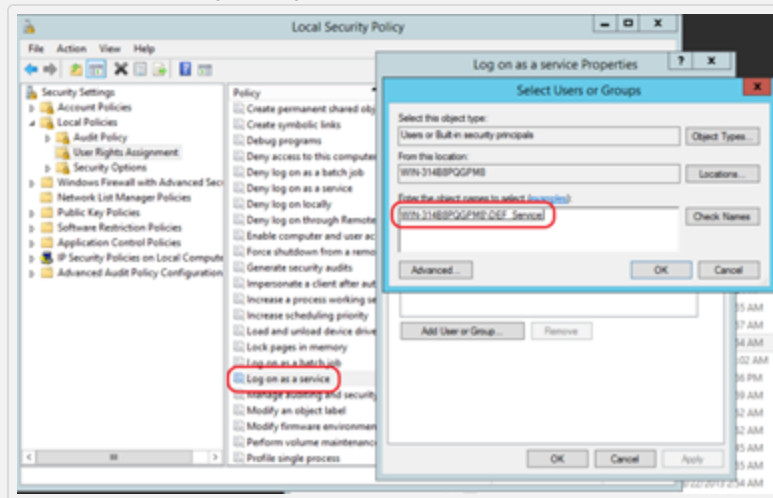
Prerequisites

- Enable the logon as a service policy for the service account:



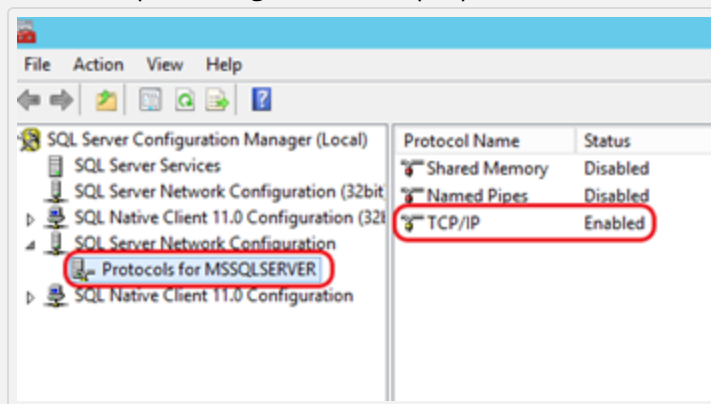
1. Go to Control PanelAdministrative ToolsLocal Security Policy.
2. Navigate to Security Settings Local Policies User Rights Assignment Log on as a service.
3. Double-click Log on as a service and click Add User or Group.
4. Enter the service account and press Check Names. If the system finds the specified account it will be displayed.

5. Exit Local Security Policy and Control Panel.



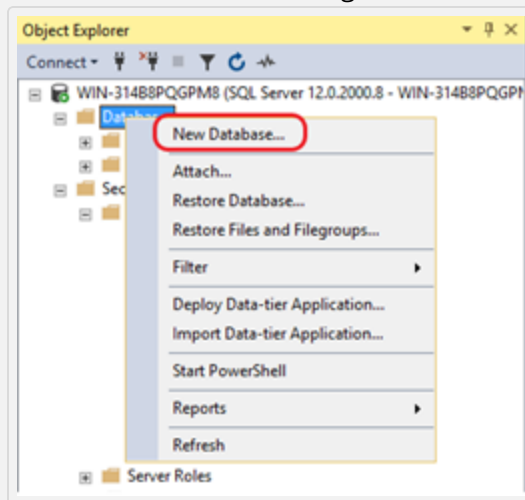
- Verify DEF is communicating with SQL Server via TCP/IP.

1. Validate the TCP/IP protocol is enabled for SQL Server.
2. Load SQL Server Configuration Manager and validate the TCP/IP protocols for the DEF SQL instance are enabled.
3. Check the port being used in the properties for TCP/IP

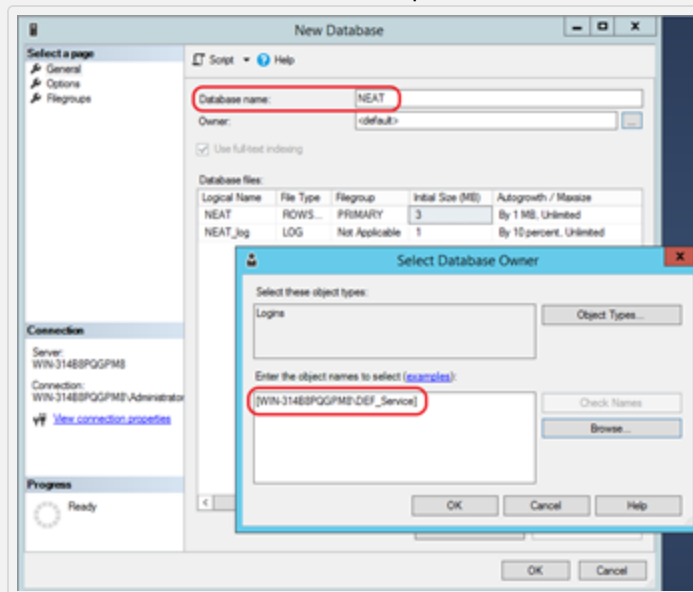


- The NEAT database can be created during the installation of the NEAT server but it is recommended to create the database prior to installation.

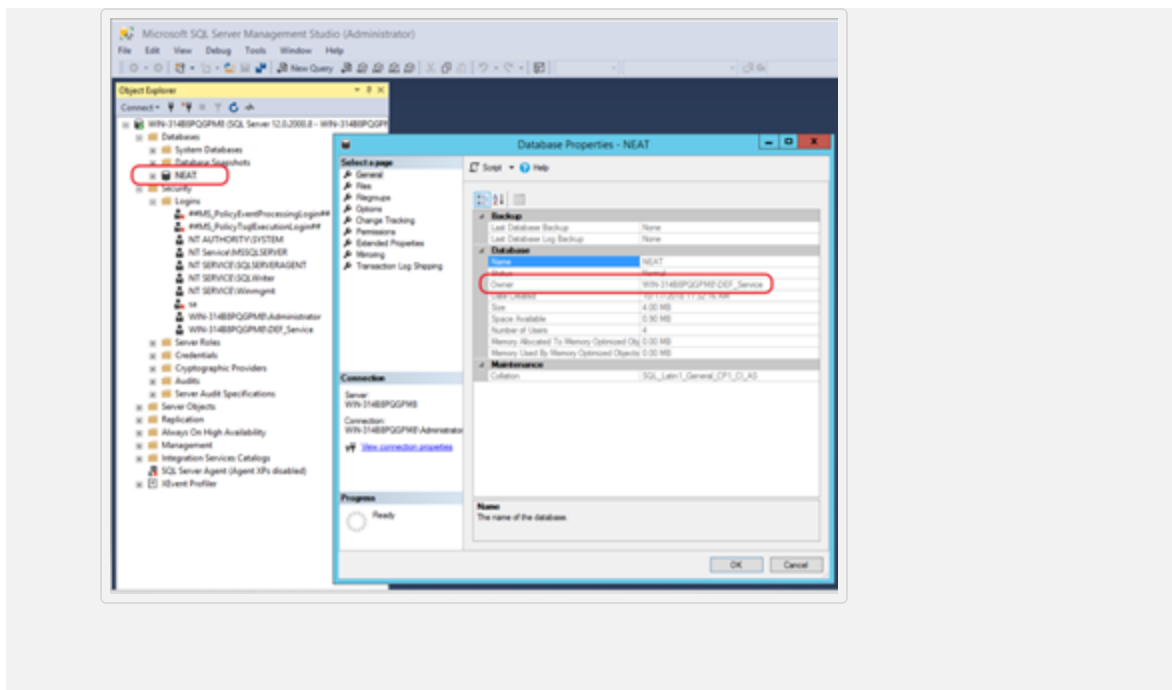
1. Select Database in SSMS, right-click and select New Database.



2. Set the Database name according to customer naming conventions and the owner to the service account. In this example the database name is NEAT.



3. Validate the NEAT database has been created and the owner is the DEF Service and exit SSMS.



Installing a NEAT Server

Installing DEF 5.x consists of two main steps:

- Unzipping the DEF directory
- Running Install.bat

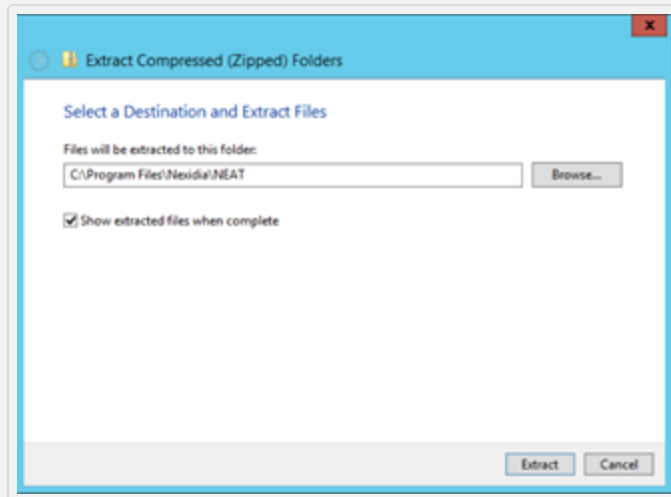
Important

Login as the DEF service account if possible.

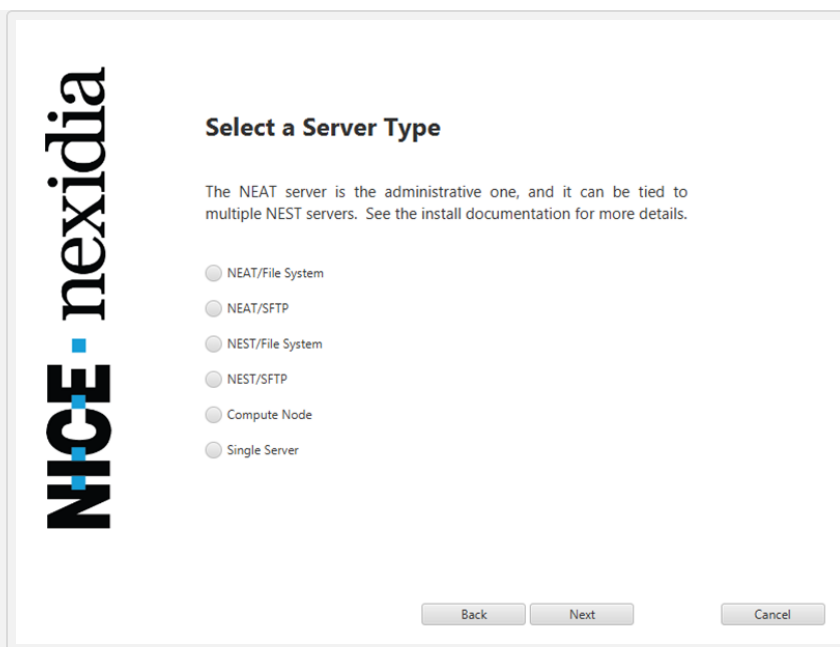
→ To install a NEAT server:

1. Create the following directory for DEF. In a customer environment the DEF should be installed to the D: drive. This example is using the C: drive due to limitations of the virtual machines being utilized

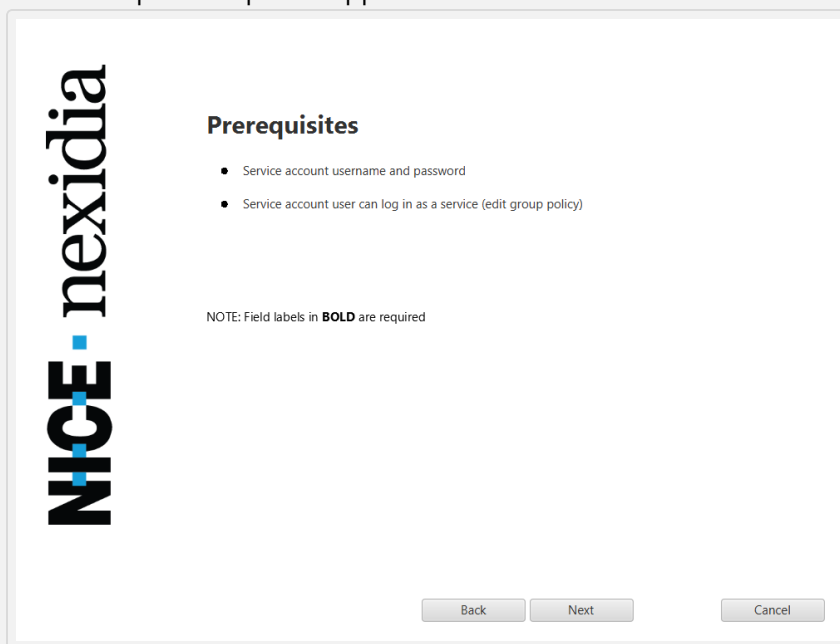
DIRECTORY	FUNCTION
C:\Program Files\Nexidia\NEAT	Install location for the DEF. During the unzip process the directory Data Exchange Framework will be installed.



2. Navigate to C:\Program Files\Nexidia\NEAT\Data Exchange Framework\install and execute Install.bat as an administrator.
You will be prompted to use the embedded JRE when the command runs. In most cases the embedded JRE is used. If the customer wishes to use a different JRE you will be prompted to provide the path to the JRE. For this example, enter Y to use the embedded JRE.
The Select a Server Type options appear.

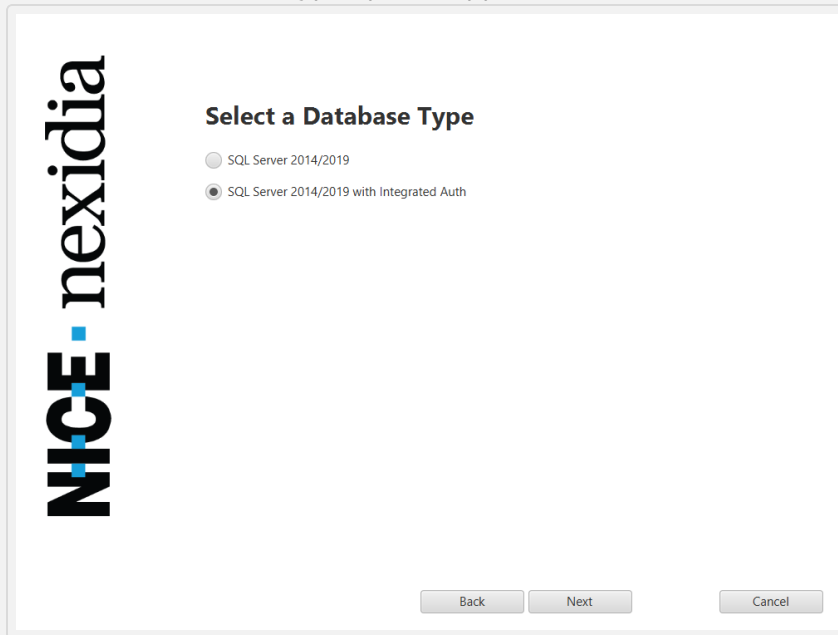


3. In the Select a Server Type window, select **NEAT/File System** and click **Next**. The Prerequisites options appear.



4. Validate all the prerequisites and click **Next**.

The Select a Database Type options appear



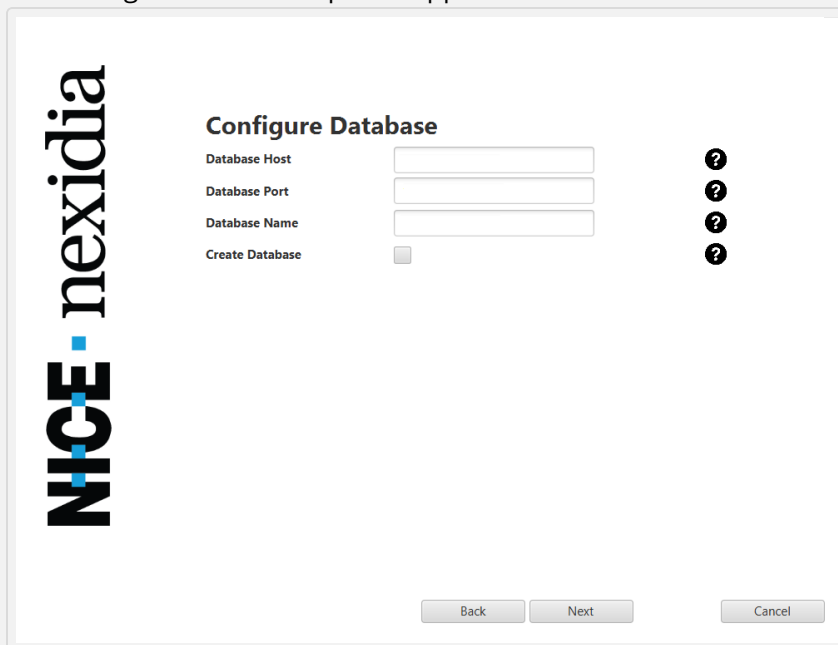
Select a Database Type

☐ SQL Server 2014/2019

☒ SQL Server 2014/2019 with Integrated Auth

Back Next Cancel

5. Select **SQL Server 2012/2014 with Integrated Auth**, and click **Next**. The Configure Database options appear.



Configure Database

Database Host

Database Port

Database Name

Create Database ☐

Back Next Cancel

6. Enter the following information:

- **Database Host:** The server on which the database resides.
- **Database Port:** The port for accessing the database. The default is 1433 for the SQL server. If the SQL service is a named instance using a dynamic port, leave this field empty.
- **Database Name:** The name of the database .
- **Create Database:** Select this checkbox to create the database if it does not already exist.

7. Click **Next**.

The installer attempts to connect to the database. If the database can be contacted, the process proceeds. If this step fails, you need to debug connectivity with the SQL server.

Configure Service

Service Account User ?

Service Account Password ?

☐ Show Password ?

Password Confirmation ?

Path to Data Folder Browse ?

Back Next Cancel

8. Enter the following information:

- **Service Account User:** The Service Account username.
- **Service Account Password:** The Service Account password.
- **Path to Data Folder:** The path to the data folder in which the Workbench files and other DEF-related data files are stored.

9. Click **Next**.

The Configure Server options appear.

Configure Server - NEAT/File

HTTP Port ?

HTTPS Port ?

Shutdown Port ?

Default Threads Per Action ?

Server retention (days) ?

DEF Internal log retention (days) ?

Audit log retention (days) ?

Disable Thread Count Auto-Tuning ☐ ?

Back Next Cancel

10. Enter the following information:

- **HTTP Port:** For connection to the NEAT user interface via the browser. The default value is 8080.
- **HTTPS Port:** For connection to the NEAT user interface via the browser. The default value is 8181.
- **Shutdown Port:** For shutdown requests from the Windows service. The default value is 8078.

- **Default Threads Per Action:** The default server-level threads per action value. It is best to set to 70% of the available processors on the NEST servers.
- **Server retention (days):** The number of days beyond which it's safe to delete the DEF internal database, contents of auto-provisioned file system resources, old rows in all working tables' rollover, and quarantine tables. Automated cleanup of these items takes place nightly. Active working table cleanup is included in the nightly run, but it uses the maximum self-reconciliation days setting at the process level, and ages out from there into the corresponding *{working table}\$Q* quarantine table. This setting then determines the aging out from there.
- **DEF Internal log retention (days):** The number of days beyond which it's safe to delete the DEF internal logs. Automated cleanup of the internal logs takes place nightly.
- **Audit log retention (days):** The number of days beyond which it's safe to delete old audit log files. Automated cleanup of the audit logs takes place nightly. Audit logs are distinct from regular logging, and only include output from those actions that require it, like the NMF Decrypt suite, or the File Mover when it's in auditing mode for specific extensions.
- **Disable Thread Count Auto-Tuning:** Select this checkbox to disable thread count auto-tuning so that all actions inherit the server-level thread count for all runs, and will need to be manually tuned for optimal performance.

Important

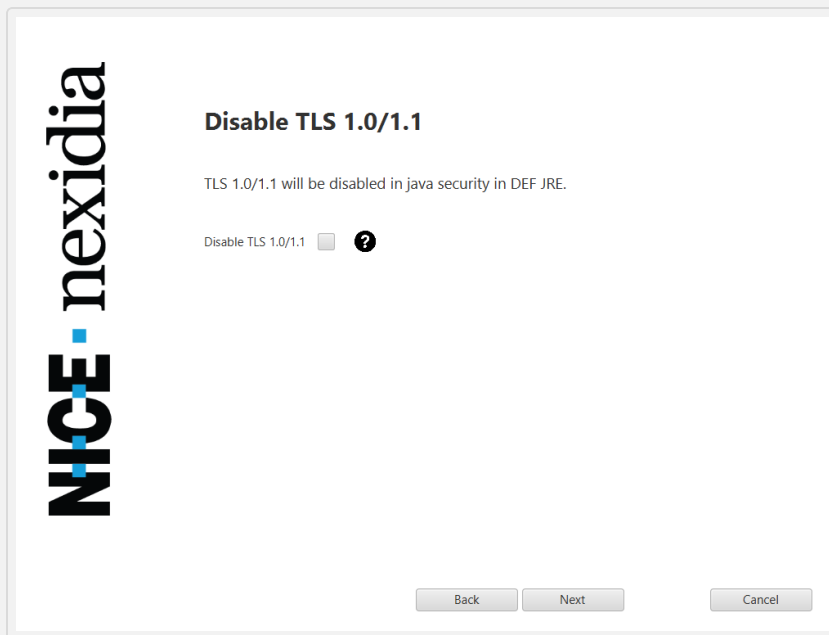
This option is not selected by default, and is only recommended for DEF installations on proxy servers or other servers where DEF contends heavily against other applications for system resources. If you select this option, a warning message appears to remind you that by selecting the option, you take responsibility for manually tuning the DEF actions.

11. Click **Next**.

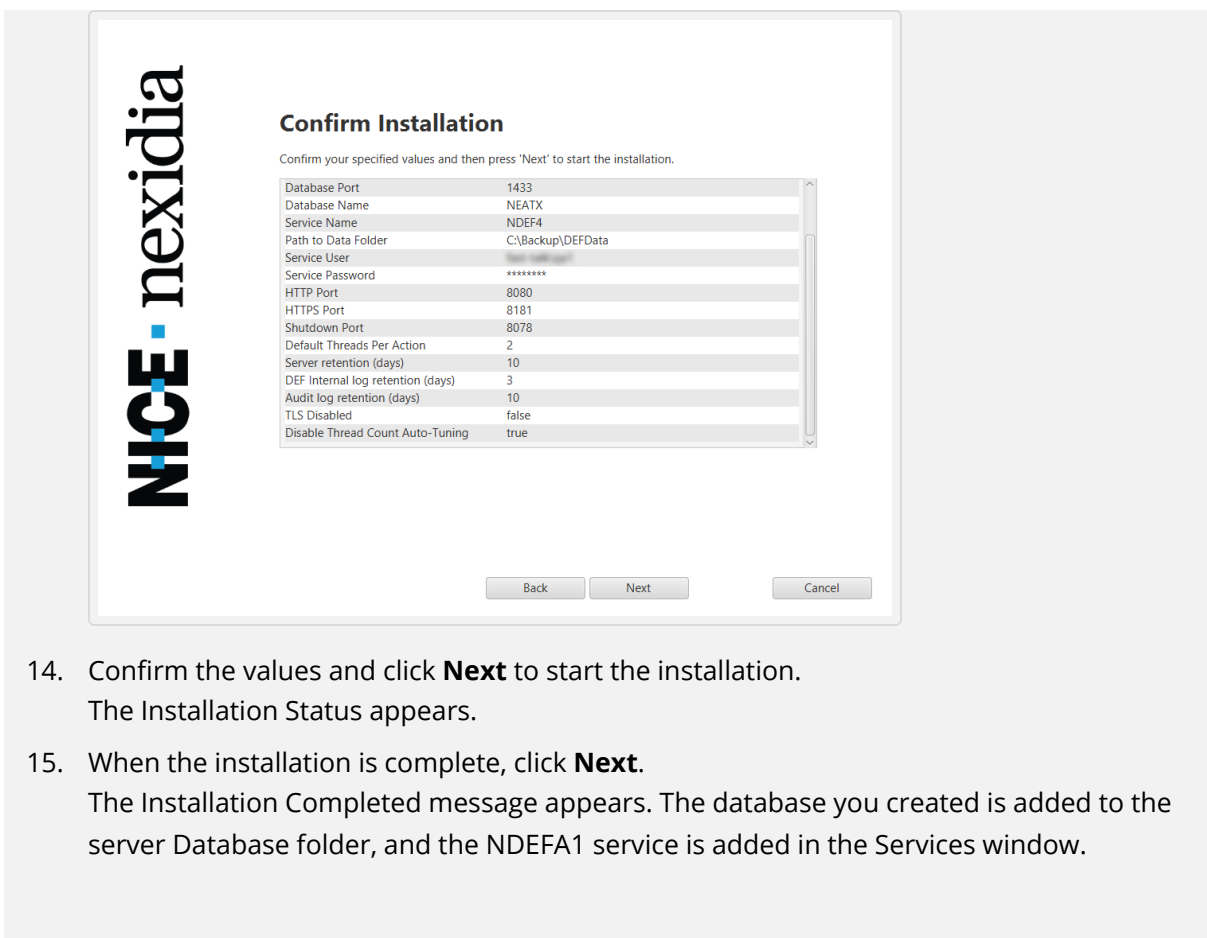
The Disable TLS 1.0/1.1 options appear.

Important

This option only appears if the embedded JRE folder doesn't already have TLS disabled for SSLv2Hello, SSLv3, TLSv1, TLSv1.1.



12. If required, select the checkbox to disable TLS 1.0/1.1.
When you select this option, the *jre/lib/security/java.security* file is updated by appending the values **SSLv2Hello**, **SSLv3**, **TLSv1**, **TLSv1.1** for the **jdk.tls.disabledAlgorithms** tag.
13. Click **Next**.
The Confirm Installation options appear.

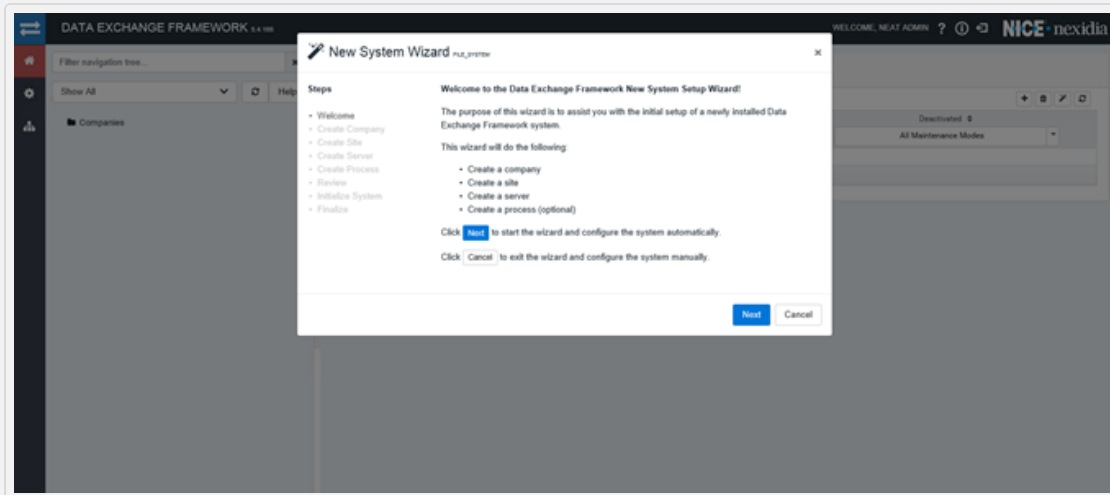


14. Confirm the values and click **Next** to start the installation.
The Installation Status appears.
15. When the installation is complete, click **Next**.
The Installation Completed message appears. The database you created is added to the server Database folder, and the NDEF41 service is added in the Services window.

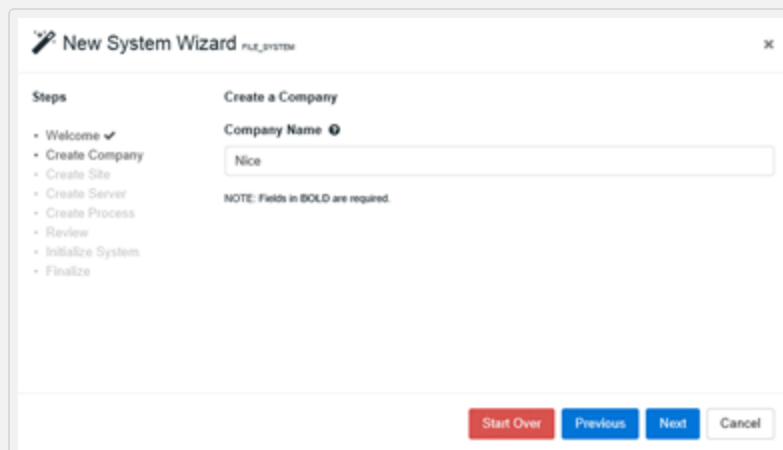
Configuring a New System

The first time the DEF NEAT page is loaded, the New System wizard runs:

1. Open your browser and enter <http://localhost:8080/#/login> or <https://localhost:8181/#/login>.



2. Follow the steps of the wizard:
 - a. Create a company.



- b. Create a site: Nexidia

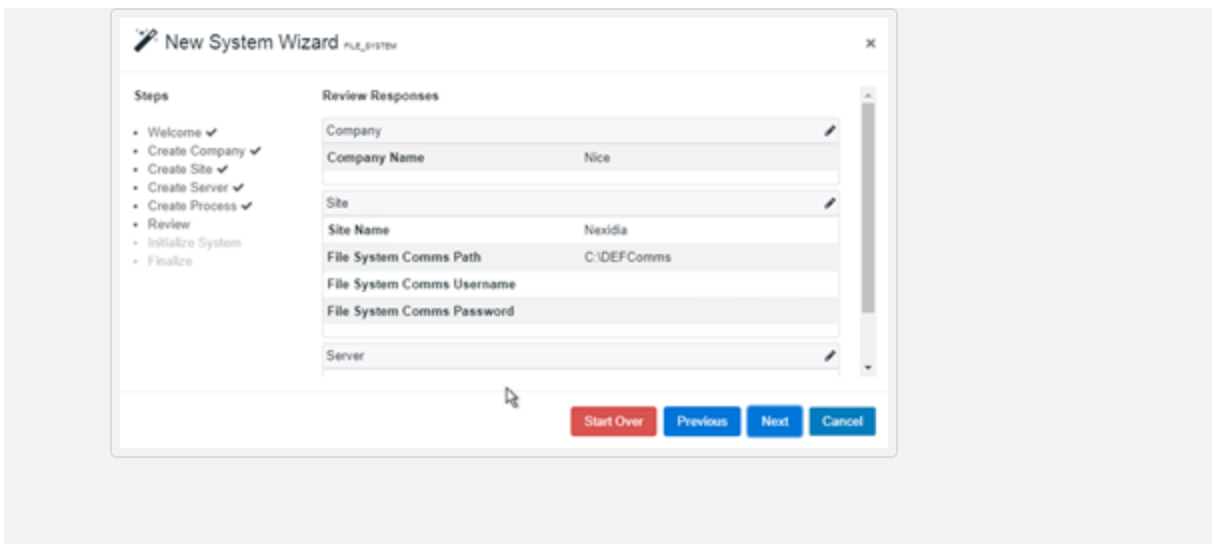
The screenshot shows the 'New System Wizard' window for a 'FILE_SYSTEM'. The 'Steps' list on the left includes: Welcome ✓, Create Company ✓, Create Site (current step), Create Server, Create Process, Review, Initialize System, and Finalize. The 'Create a Site' section contains four input fields: 'Site Name' with 'Nexidia' entered, 'File System Comms Path' with the placeholder 'Enter a valid file system communications path', 'File System Comms Username' with the placeholder 'Enter a valid file system communications path username', and 'File System Comms Password' with the placeholder 'Enter a valid file system communications path password'. At the bottom are buttons for 'Start Over', 'Previous', 'Next', and 'Cancel'.

Since this is a NEAT File System server, you are prompted for the network share for NEAT/NEST communications. If the network share has not been created, it must be created before proceeding further.

- c. Create a server: ATL

The screenshot shows the 'New System Wizard' window for a 'FILE_SYSTEM' at the 'Create a Server' step. The 'Steps' list on the left includes: Welcome ✓, Create Company ✓, Create Site ✓, Create Server (current step), Create Process, Review, Initialize System, and Finalize. The 'Create a Server' section contains two input fields: 'Server Name' with 'ATL' entered and 'Communications Directory' with 'ATL' entered. A note at the bottom states 'NOTE: Fields in BOLD are required.' At the bottom are buttons for 'Start Over', 'Previous', 'Next', and 'Cancel'.

- d. Enter the server name and a name for the communication directory. It is recommended that the server name and communication be the same.
- e. To finish, review your entries and click **Next**.



NEST Server Installation

Prerequisites

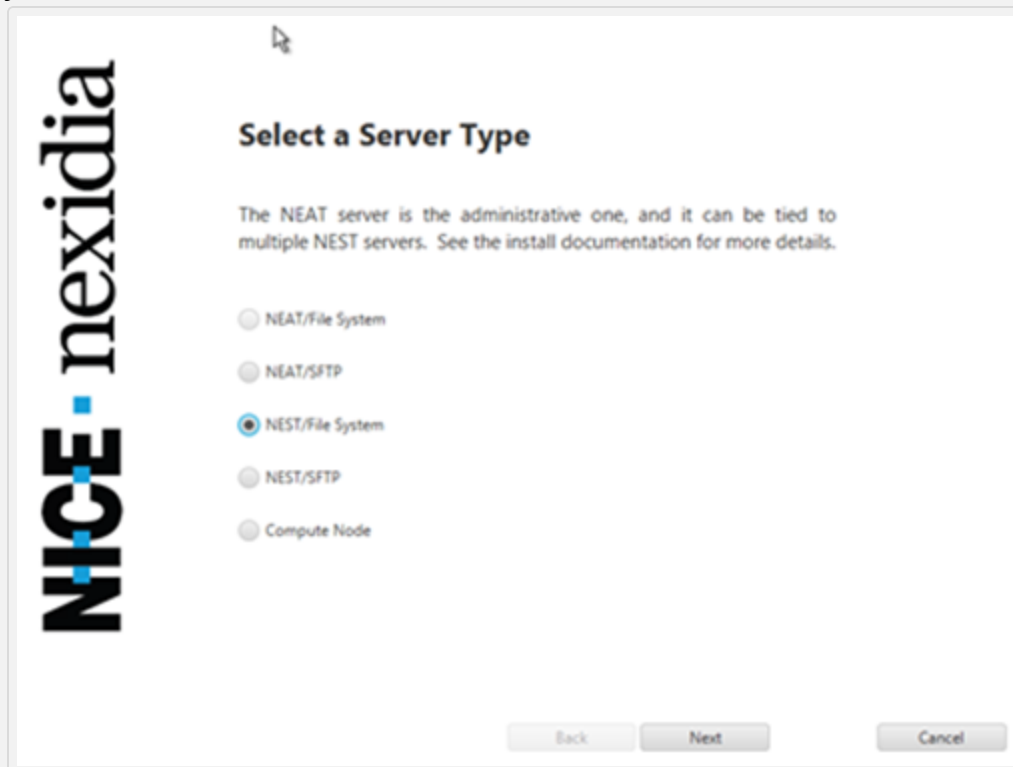
- Enable the logon as a service policy for the service account on the NEST server.
- Verify DEF is communicating with SQL Server via TCP/IP. Validate the TCP/IP protocol is enabled for SQL Server.
- The NEST database can be created during the installation of the NEST server but it is recommended to create the database prior to installation.
- Set the database name according to customer naming conventions and the owner to the service account. In this example, the database name is NEST.

Installing a NEST Server

1. Create the following directory for DEF. In a customer environment the DEF should be installed to the D: drive. This example is using the C: drive due to limitations of the virtual machines being utilized

DIRECTORY	FUNCTION
C:\Program Files\Nexidia\NEST	Install location for the DEF. During the unzip process the directory Data Exchange Framework will be installed.

- Unzip the DataExchangeFramework-Install-5.x.zip file and point to C:\Program Files\Nexidia\NEST as the destination
- Navigate to C:\Program Files\Nexidia\NEST\Data Exchange Framework\install and execute Install.bat as an administrator.
You will be prompted to use the embedded JRE when the command runs. In most cases the embedded JRE is used. If the customer wishes to use a different JRE you will be prompted to provide the path to the JRE. For this example, enter Y to use the embedded JRE



- In the Server Type, select NEST/File System and click **Next**. The Prerequisites window displays.

5. Validate all the prerequisites and click **Next**. The Database type window displays.
6. Select SQL Server 2012/2014 w/Integration, and click **Next**. The Configure Database window displays.
 - a. Enter the Database Host, Port and Name.
 - b. Enter the FQDN for the SQL server. The database port may be left blank if the SQL instance is enabled for dynamic port.
 - c. Enter the name of the database that was created before starting the installation or to be created during the installation.
7. Click **Next**.
The installer attempts to connect to the database. If the database can be contacted, the process proceeds. If this step fails, you need to debug connectivity with the SQL server.
8. In the Configure Service window, enter the following information:

SETTING	RECOMMENDED VALUE
Service User Account	The name of the service account provided by the customer
Service User Acct Pwd	Password provided by customer for the DEF service account
Enter full path to data directory	C:\Program Files\Nexidia\NEST\Data Exchange Framework\data Note: Do not use the C: on a customer install. Use the D: or the drive specified by the customer

The installer attempts to validate the information provided. Any failures are trapped and displayed so they can be corrected. In this example, the service account password was entered incorrectly

9. Click **Next**. The Configure Server window opens.

Configure Server - NEST/File

NEST Server ID: 117

File Comms Path: C:\DEFComm\ATL Browse

File Path Username (CIFS): File path username

File Path Password (CIFS): File path password

Shutdown Port: 8078

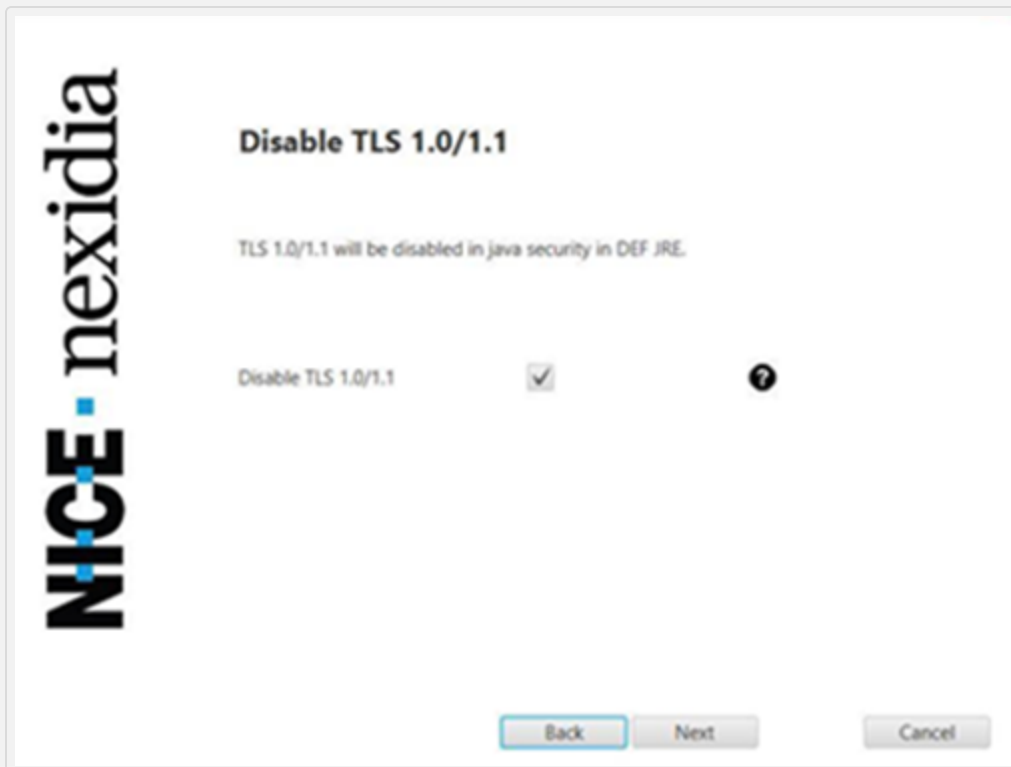
NOTE: CIFS username and password are optional

Back Next Cancel

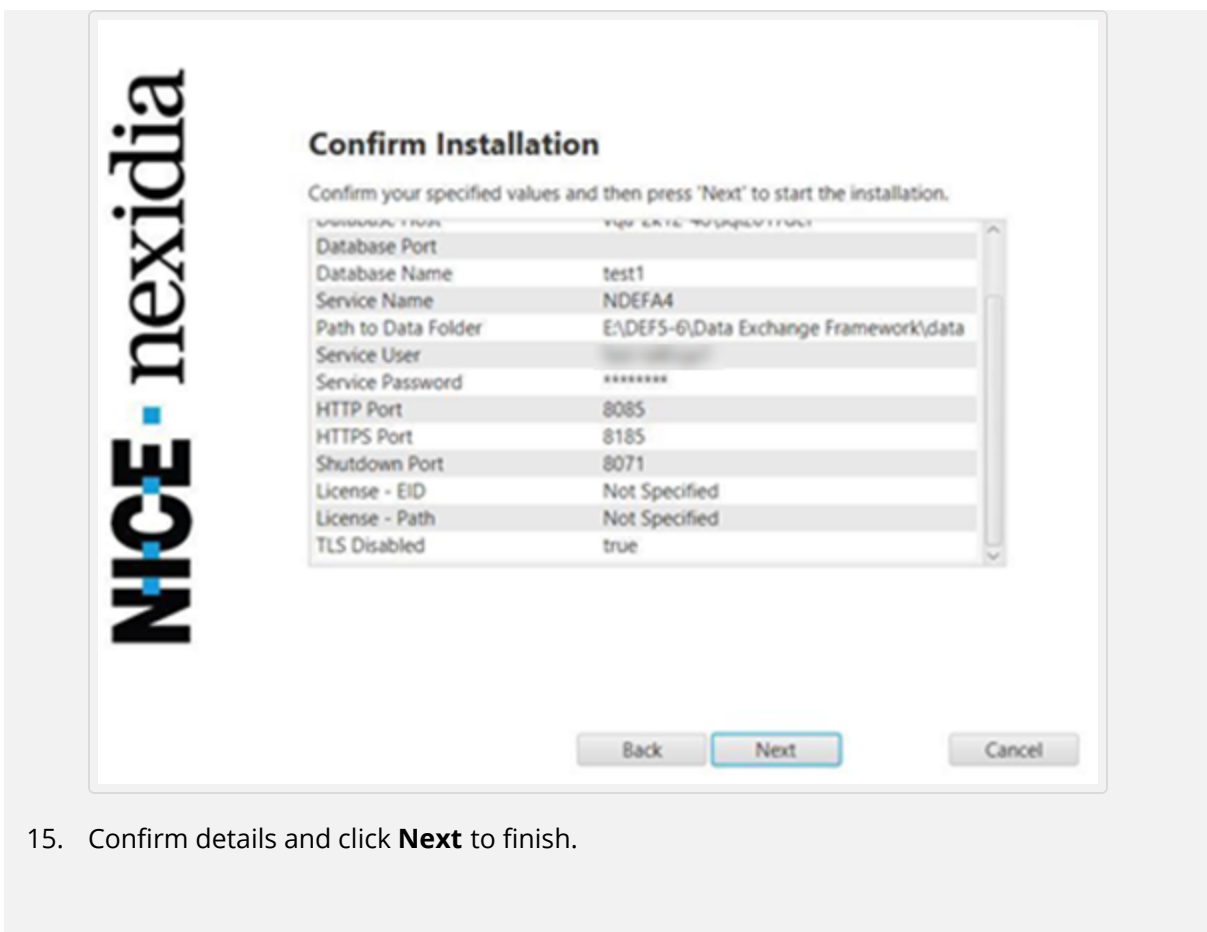
10. Manually create the NEST server on the NEAT server. See [Manually Creating a NEST Server on the NEAT Server](#) on page 62.
11. Enter details as follows:
 - **NEST Server ID**: From the NEAT NEAT UI server properties tab
 - **File Path**: This is the path over which NEAT/NEST communications will happen. For NEST servers, this should be a concatenation of the site and server communication paths assigned in the NEAT user interface
 - (Optional) **File Path Username** (CIFS only)
 - (Optional) **File Path Password** (CIFS Only)
 - **Shutdown Port**: This is the port the Windows service will use to communicate shutdown requests to the DEF
12. Click **Next**. The Disable TLS 1.0/1.1 window opens.

Important

If External JRE is selected on the command prompt at the start of the install, then this window does not display. The window displays if embedded JRE folder doesn't already have the TLS disabled for SSLv2Hello, SSLv3, TLSv1, TLSv1.1. If these are already included in the embedded jre folder, then the TLS screen does not display.



13. Select the checkbox to disable TLS 1.0/1.1.
If Disable TLS is selected, it updates the `jre/lib/security/java.security` file by appending the values: `SSLv2Hello`, `SSLv3`, `TLSv1`, `TLSv1.1` for the tag: `jdk.tls.disabledAlgorithms`
14. Click **Next**. The Confirm Installation window opens.



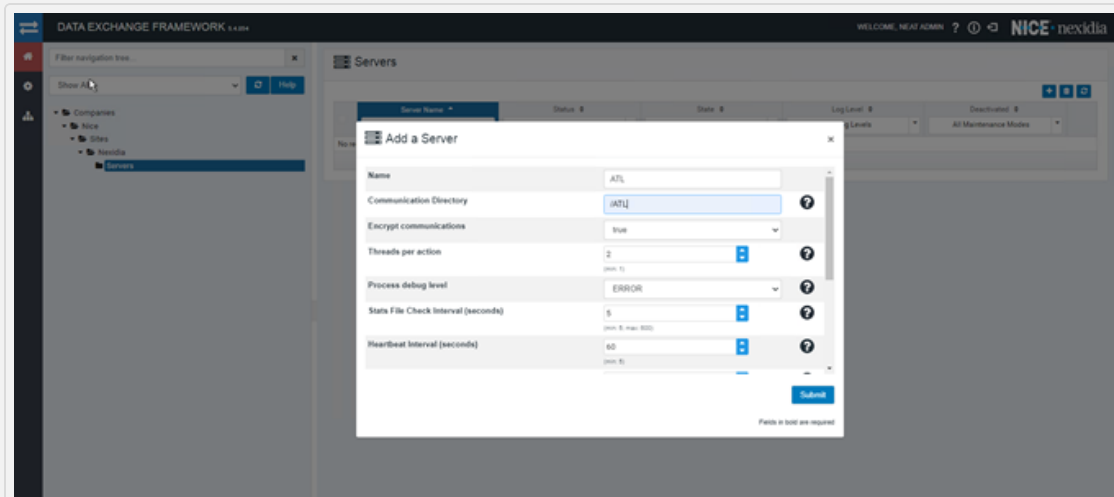
15. Confirm details and click **Next** to finish.

Manually Creating a NEST Server on the NEAT Server

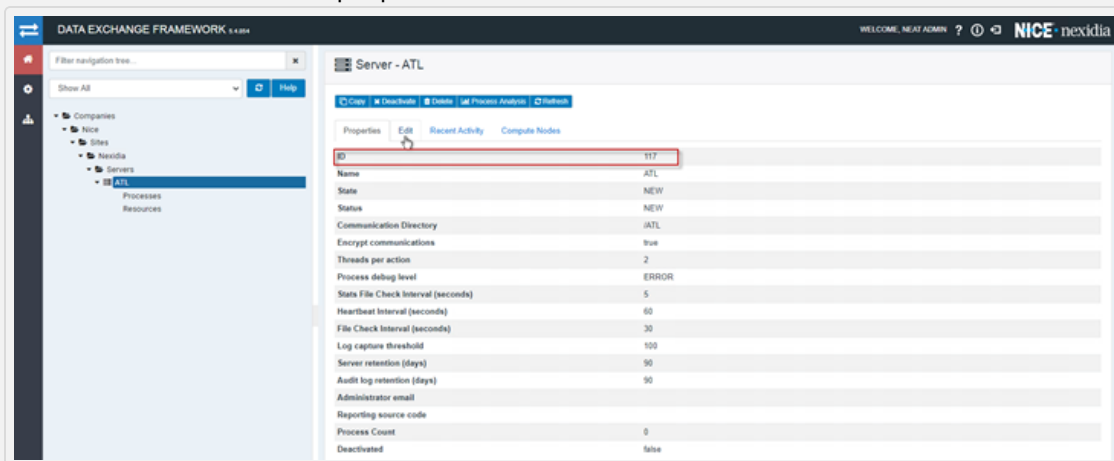
A NEST server must be created in the NEAT UI before the NEST server can be installed. It can be created when running the NEAT server New System Wizard, or it can be created manually.

→ To create manually

1. Add a server in the NEAT UI.



2. Enter a name and communication path for the server. It is recommended that the communication directory be the same name as the NEST server
3. Configure the NEST server with a Server ID and a communication path. The NEST server ID can be found under the properties tab if the NEST server.



4. On completion, return to the Configure Server step in the UI.

MULTI-SERVER INSTALLATION USING CONFIG FILE

In some instances, you can install multiple remote NEAT/NEST servers based on a config file. The config file has default values for each parameter. You can enter the required values for the relevant parameters directly into the config file and the same configuration can be used for multiple NEAT/NEST servers. This config file also handles the cluster failover manager.

→ To access the config file:

1. In the location where you saved the current **DataExchangeFramework** installation package, navigate to **Install > MultiServersInstall > config**.
2. Open the **batchInstallTemplate.properties** file in a relevant editing tool and in each section, enter parameters for the NEAT and NEST servers.
3. Follow the instructions inside the file for entering the required information.
4. Copy and paste all the parameters for each of the NEST/NEAT servers required and adjust the configuration values for the specific NEST/NEAT server, such as `HOST_NAME`, `INSTALL_DIRECTORY`.
5. For cluster installation, enter all relevant values for each of the properties related to the Cluster environment; name, role and other relevant configuration requirements.
6. When you are finished, save the properties file.
7. Open a Command prompt and change directory to the **Install>MultiServerInstall** directory.
8. Run **MultiServersInstallation.bat**. This opens an Installer window and prompts for any required passwords that were not entered in the properties file.
9. Enter the passwords as prompted and click **Enter**. All the values entered in the properties file are validated and if there are any validation errors, they are prompted here.
10. Close the popup errors and the Installer window and fix the errors as needed.
11. Relaunch the installer from the command prompt. Once the validations pass, the confirmation window opens with all the NEAT and NEST instances included in the properties file.

12. Click **Next**. Installation status window shows each instance being installed.
13. Once all the instances are completed, click **Next** to close the installer window.

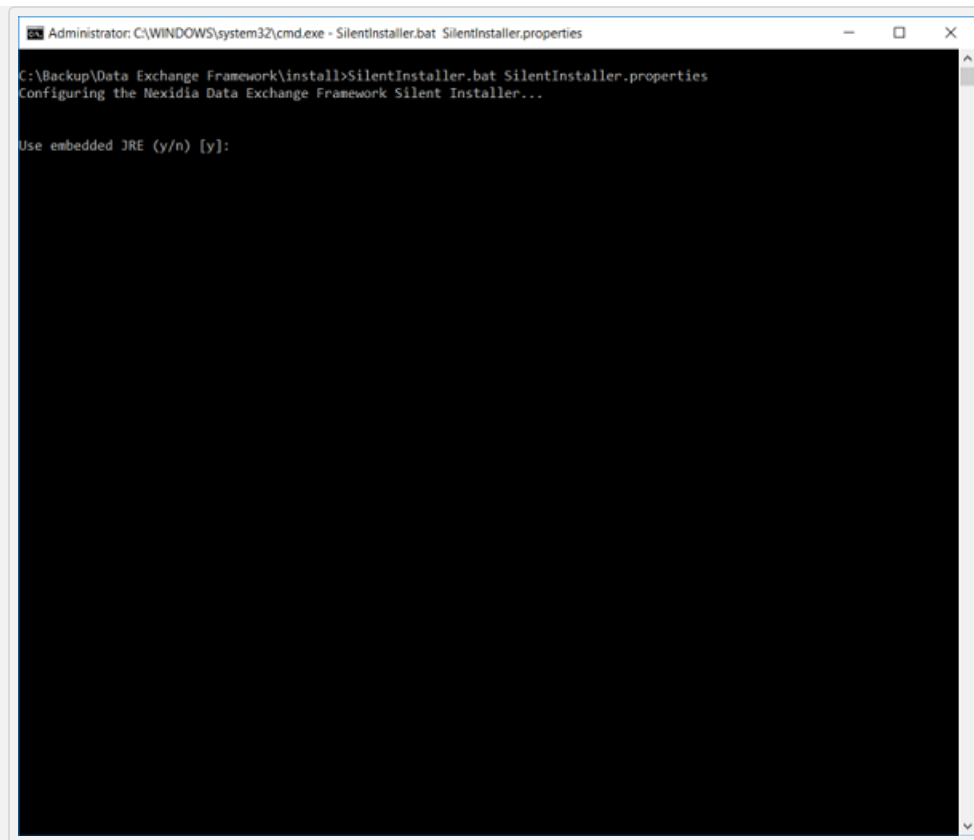
SILENT INSTALLATION

The Data Exchange Framework silent installation is a command line utility. These files are delivered with your installation package:

- `SilentInstaller.bat` - Auto-installs the application. The configuration selections in the `SilentInstaller.properties` file are used by this file.
- `SilentInstaller.properties` - Contains the configuration options that you will specify for your installation. All of the options that are available in the regular installation process are in this file. Items that are in angle brackets <> must be updated. The configuration options for each installation type are included in this file.

→ To perform a silent installation:

1. Unzip the *DataExchangeFramework-Install-X.X.zip* file to the **root\Data Exchange Framework\install** directory.
2. In the *SilentInstaller.properties* file, specify the options for the server installation. The file provides details about the values that need to be defined.
3. From the command line, change directories to the **root\Data Exchange Framework\install** directory.
4. From the command line, run the silent installer by typing:
`.SilentInstaller.bat SilentInstaller.properties`
5. You are prompted to specify if you want to use the embedded JRE.
 - Press **Enter** to select the default option of "y" to use the embedded JRE.
 - Enter "**n**" if you want do not want to use the embedded JRE. If you select this option, you will need to have JAVA_HOME in your environment. If you have not set up your JAVA_HOME, you are asked to enter the path to the JRE.

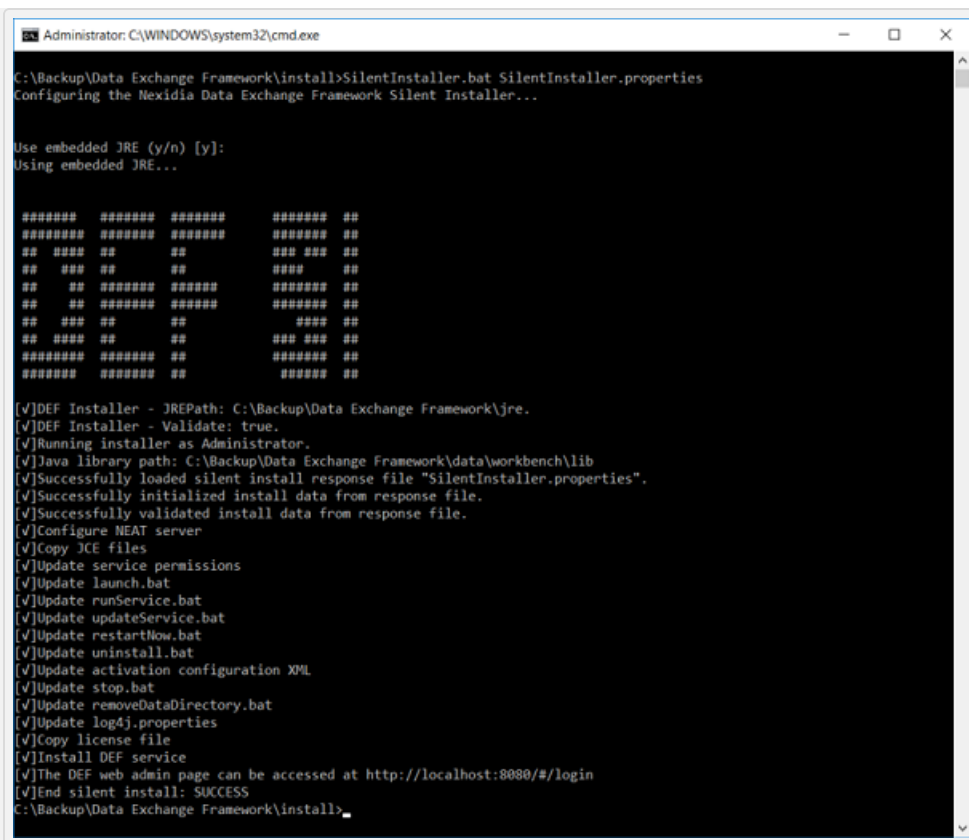


```
Administrator: C:\WINDOWS\system32\cmd.exe - SilentInstaller.bat SilentInstaller.properties

C:\Backup\Data Exchange Framework\install>SilentInstaller.bat SilentInstaller.properties
Configuring the Nexidia Data Exchange Framework Silent Installer...

Use embedded JRE (y/n) [y]:
```

Validations are performed and you are notified if the installation failed or succeeded. If it failed, the causes for the failure are listed.



```
Administrator: C:\WINDOWS\system32\cmd.exe

C:\Backup\Data Exchange Framework\install>SilentInstaller.bat SilentInstaller.properties
Configuring the Nexidia Data Exchange Framework Silent Installer...

Use embedded JRE (y/n) [y]:
Using embedded JRE...

#####
#####
##      ##      ##      ##      ##
##      ##      ##      ##      ##
##      ##      ##      ##      ##
##      ##      ##      ##      ##
##      ##      ##      ##      ##
##      ##      ##      ##      ##
##      ##      ##      ##      ##
#####

[V]DEF Installer - JREPath: C:\Backup\Data Exchange Framework\jre.
[V]DEF Installer - Validate: true.
[V]Running installer as Administrator.
[V]Java library path: C:\Backup\Data Exchange Framework\data\workbench\lib
[V]Successfully loaded silent install response file "SilentInstaller.properties".
[V]Successfully initialized install data from response file.
[V]Successfully validated install data from response file.
[V]Configure NEAT server
[V]Copy JCE files
[V]Update service permissions
[V]Update launch.bat
[V]Update runService.bat
[V]Update updateService.bat
[V]Update restartNow.bat
[V]Update uninstall.bat
[V]Update activation configuration XML
[V]Update stop.bat
[V]Update removeDataDirectory.bat
[V]Update log4j.properties
[V]Copy license file
[V]Install DEF service
[V]The DEF web admin page can be accessed at http://localhost:8080/#/login
[V]End silent install: SUCCESS
C:\Backup\Data Exchange Framework\install>
```

SilentInstaller.properties File

An example SilentInstaller.properties file is shown below.

```
#  
# DEF silent installer properties file  
#  
# NOTE: Items in angle brackets <> need to be changed  
#  
# Select server type  
# Valid values: NEAT/File/ NEAT/SFTP, NEST/File, NEST/SFTP, ComputeNode, Single Server  
ServerType=NEAT/File  
# Select database type  
# Valid values: Integrated Auth, Password  
SqlServerAuthType=Integrated Auth  
# Configure database  
DefDatabaseHost=<def database host name>  
DefDatabasePort=1433  
DefDatabaseName=<def database name>  
DefDatabasePassword=  
DeFDatabaseUsername=  
# DefDatabaseCreate values are true or false  
DefDatabaseCreate=  
#DefDatabaseLogPath: Path to the DB Log directory to be used when creating DEF database if it  
doesn't exist  
DefDatabaseLogPath=  
#Default server-level threads per action value. It is best to set to 70% of available processors on  
the NEST servers.  
#If the server type is single server change this property to 2  
DEFAULT_THREADS_PER_ACTION=4
```

#The number of days beyond which it's safe to delete DEF internal database, the contents of auto-provisioned file system resources, and old rows in all working tables' rollover and quarantine tables. Automated cleanup of these items happens nightly. Note that active working table cleanup is also included in the nightly run, but it uses the maximum self-reconciliation days setting at the process level, and ages out from there into the corresponding '{working table}\$Q' quarantine table. This setting then determines the aging out from there.

#If the server type is single server change this property to 10

SERVER_RETENTION=90

#The number of days beyond which it's safe to delete DEF internal logs. Automated cleanup of these items happens nightly.

#If the server type is single server change this property to 3

LOG_RETENTION=14

#The number of days beyond which it's safe to delete old audit log files. Automated cleanup of these items happens nightly. Note that audit logs are distinct from regular logging, and include only output from those actions that require it, like the NMF Decrypt suite, or the File Mover when it's in auditing mode for specific extensions.

#If the server type is single server change this property to 10

AUDIT_RETENTION=90

#Disables thread count auto tuning for the process. set to true to disable that. By default auto tuning is enabled.

#For Single Server service type set this to true

#Note that this option is recommended for DEF installs on proxy servers, or other servers where DEF contends heavily against other applications for system resources.

#Disabling auto-tuning means that all actions will inherit the server level thread count for all runs, and this single number will almost never optimize performance for each action;

#so action-level fine-tuning of thread counts will be required in many cases.

#Some actions may be I/O-bound and could use more, and others may be CPU-bound and risk being oversubscribed.

#By continuing with this option, you accept responsibility for tuning all DEF actions to appropriate settings for the environment,

#and considering their interaction with other actions in other processes running in parallel, in order to optimize performance and not risk oversubscribing the CPU.

DISABLE_THREAD_COUNT_AUTO_TUNING=

Configure service

DefServiceAcctUser=<domain name>\\<username>

DefServiceAcctPassword=<service account password>

DefDataFolder=C:\\Backup\\DEFData

Configure server - NEAT/File, NEAT/SFTP

BaseHttpPort=8080

BaseHttpsPort=8181

BaseShutdownPort=8078

Configure server - Nest/File

NESTFileServerId=

NESTFileFileCommsPath=

NESTFileFilePathUsername=

NESTFileFilePathPassword=

NESTFileShutdownPort=8078

Configure server - Nest/SFTP

NESTSFTPServerId=

NESTSFTPSFTPHost=

NESTSFTPPort=22

NESTSFTPCommsPath=

NESTSFTPUsername=

Valid values: Password, Key Based

NESTSFTPPasswordType=Password

NESTSFTPPassword=

NESTSFTPKeyPath=

NESTSFTPKeyPassPhrase=

NESTSFTPShutdownPort=8078

Configure server - Compute Node

CNInstallFolder=\\<host name>\c\$\Program Files\Nexidia\Data Exchange Framework

CNDataFolder=\\<host name>\c\$\Program Files\Nexidia\Data Exchange Framework\data

CNLocalDataFolder=C:\Backup\DEFData

CNShutdownPort=8078

#TLS_DISABLED disable TLS for Internal JRE only, can be one of the following: (True, False) and by default is True. It will be ignored if it is not Internal JRE

TLS_DISABLED=true

Migrating DEF 3.5 to DEF 5.x

This section details the process for installing DEF 5.x on a server with an existing DEF 3.5 installation. Migration from DEF 3.5 to DEF 5.x requires installation of the newer version rather than an upgrade. It is not necessary to uninstall 3.5 to install any 5.x version.

Important

Although this process is available, it is not recommended to install DEF 3.5 and DEF 5.x on the same server.

REQUIRED DEF 3.5 INFORMATION

1. Open the `ndef.properties` file from the DEF 3.5 installation. This file is located in the DEF install directory: **"D:\Program Files\Nexidia\Data Exchange Framework\etc"**.
2. In the `ndef.properties` file, locate the ports that are being used for HTTP, HTTPS and Shutdown:
 - `ndef.server.type=NEAT`
 - `ndef.communication.mode=SINGLE_SERVER`
 - `nx.workbench.product.key=vIA/WdV+FXFqIBHDhKF3vWI0uu05QScNcwBFosTcBk/xKL/5zSk0Zj/VhN9L46XJ| | | zFfXD3yNMykXe0btY725Ew==`
 - `ndef.database.username=developer`
 - `ndef.database.pwd=3A0+mBUcKq1Pvcu6sYaviQ== | | | dv9bDnJ03BaK3pEtSBT/Mw==`
 - `ndef.shutdown.listen.port=8078`

- `ndef.admin.https.keystore.pwd=QSMkmkPr5joTN7+/VTxhfQ== || atjT8rNf5pkpXUfRS6YbQA==`
- `user.home=E:\\Program Files\\Nexidia\\Data Exchange Framework\\data`
- `ndef.admin.http.port=8080`
- `ndef.admin.https.port=8181`

INSTALLING DEF 5.X

This process applies to all DEF 5.x versions. The process starts after DEF Installation, and extraction of the DEF application to an installation directory.

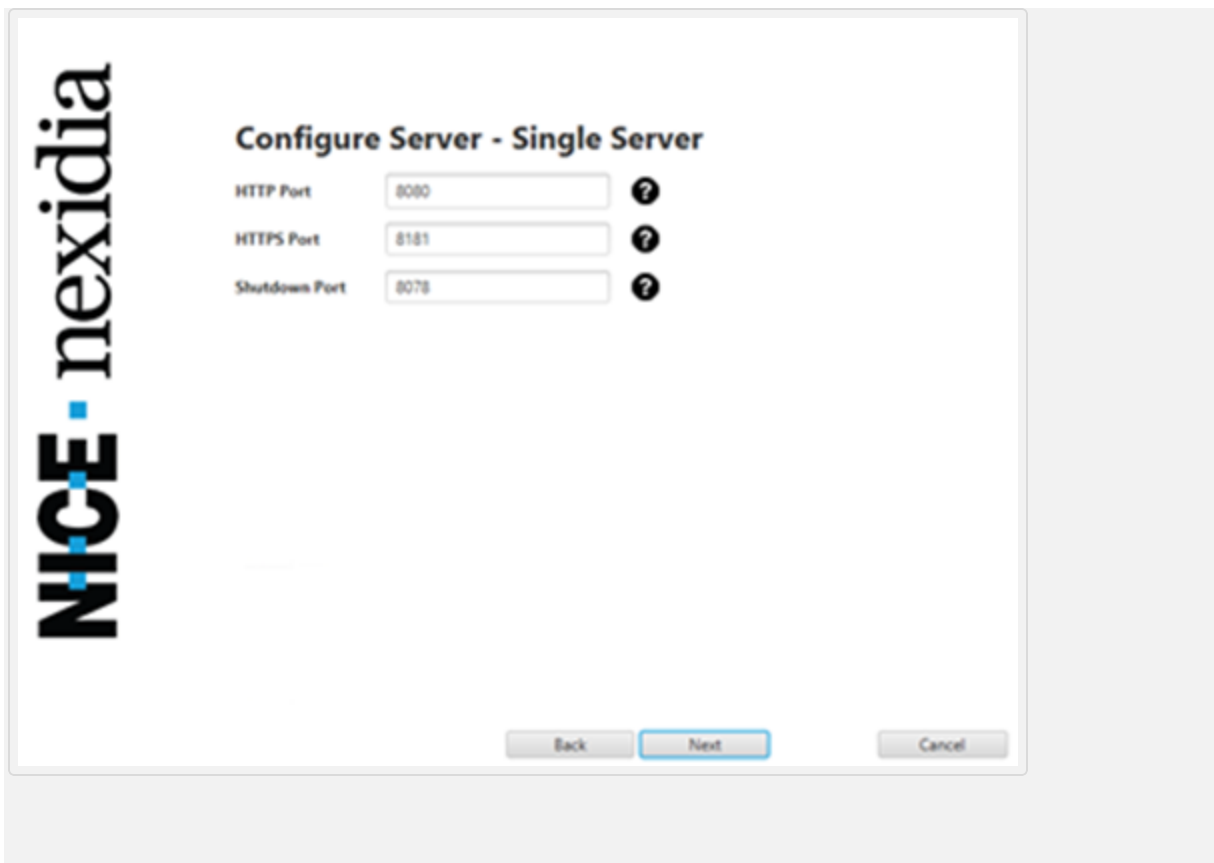
Important

DEF cannot be installed to the same folder as DEF 3.5.

1. Execute the Install.bat file.
2. In the Configure Server dialog, change the port assignments so they do not conflict with DEF 3.5.
3. Use "netstat -ano" at a command prompt to list all active ports. You must select three ports that are not in use. "netstat -ano | findstr "xxxx"" can be used to test ports to see if they are in use. The recommended ports are:
 - HTTP Port: 8081
 - HTTPS Port: 8182
 - Shutdown Port: 8076

Important

You must validate these ports are available using the "netstat -ano | findstr "xxxx"" command.



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Configure Server - Single Server

HTTP Port	8080	?
HTTPS Port	8181	?
Shutdown Port	8078	?

Back Next Cancel

PORTING PROCESSES FROM DEF 3.5 TO DEF 5.X

Process JSON files can be created in DEF 3.5 and then imported into DEF 5.x. Any SQL code that is used in CustomSQL, DB-to-CSV or other actions must be tested to validate it is compatible with the SQL version used by DEF 5.x.

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Sentinel Monitoring

A RESTful endpoint is available that captures several key metrics across servers that are part of the deployment and delivers them in JSON back to the caller.

- `http://hostName:8080/NEAT/def/api/v1/application/sentinel`

This call requires authentication against an authentication endpoint first, and then the returned token is used in subsequent requests using *token* as the key.

- `http://hostName:8080/NEAT/def/api/v1/login/authenticate`

RETURNED METRICS

These metrics and their corresponding Sentinel IDs are returned. They reflect counts from the past hour.

METRIC	SENTINEL ID
Records imported into the Data Exchange Framework	2001
Records processed successfully	2002
Records failing (retrying)	2003
Records failed (gave up)	2004
Records delivered to Interaction Analytics	2005
Records decrypted	2006

The call breaks down the statistics by server (in server ID order), and maps **all Worker Servers to component ID 1501**.

SAMPLE JSON OUTPUT

This is a sample of a full JSON output format. The hostname fields populate with the server names as configured in the DEF. Any needed modifications must occur there.

The call triggers the Worker server to gather the statistics. The first request shows zero results. Results display as the servers report in. Worker Servers gather the metrics every 10 minutes and send them back to the Admin server, where they are available for reporting.

```
{
  "data": [
    {
      "metricId": 2001,
      "metricValue": "102",
      "componentId": 1501,
      "hostname": "vqa-12r2-1210-3"
    },
    {
      "metricId": 2002,
      "metricValue": "100",
      "componentId": 1501,
      "hostname": "vqa-12r2-1210-3"
    },
    {
      "metricId": 2003,
      "metricValue": "1",
      "componentId": 1501,
      "hostname": "vqa-12r2-1210-3"
    },
    {
      "metricId": 2004,
```



```
"metricValue": "1",  
"componentId": 1501,  
"hostname": "vqa-12r2-1210-3"  
},  
{  
  "metricId": 2005,  
  "metricValue": "100",  
  "componentId": 1501,  
  "hostname": "vqa-12r2-1210-3"  
}  
]  
}
```

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Upgrading Data Exchange Framework

The Cumulative Updater tool is used to upgrade the Data Exchange Framework. You can choose to upgrade individual servers, as described in [Upgrading an Individual Server](#) on page 87, upgrade multiple servers at the same time from one location, as described in [Upgrading Multiple Servers](#) on page 94 or upgrade a clustered environment, as described in [Upgrading a Clustered DEF Environment](#) on page 96.

Before you run the Cumulative Updater tool, you can validate that the classes in your custom action JAR file (*actions.jar*) are compatible with the Data Exchange Framework version to which you are upgrading, as described in [Validating Custom Actions](#) on the next page. Validating the custom actions ahead of time helps avoid scenarios in which a planned upgrade is blocked because of incompatible custom actions.

Warning

When you upgrade your DEF installation, the Administration (NEAT) server and all Worker (NEST) servers must be upgraded to the same DEF version.

Important

If you are upgrading from a version prior to 5.0, you must install a new DEF instance with a dedicated database (separate from the previous release), and then recreate the processes there.

VALIDATING CUSTOM ACTIONS

Before you run the Cumulative Updater, you can validate that the classes in your custom actions JAR file (*actions.jar*) are compatible with the Data Exchange Framework version to which you are upgrading. The validation tool checks that your custom actions file is compatible with Java 11, as well as whether any custom actions can be replaced by core actions.

You can validate your custom actions from the command prompt or from a dedicated user interface.

→ To validate custom actions from the command prompt:

1. Navigate to the *CumulativeUpdateTool-X.X.zip* file and extract it to a local folder.
2. Navigate to the **validateCustomActionInCU > validateCustomActionViaConfig > config** folder.
3. Open the *batchValidateCustomActionInCUTemplate* properties file.
4. Specify where each instance of DEF to be validated is located. You can add as many lines as you need based on how many servers you have. You must include these values:
 - **DOMAIN_NAME**: The server name.
 - **INSTALL_DIRECTORY**: The directory where the Data Exchange Framework is installed. For example, *c:\Program Files\Nexidia*. Do not include Data Exchange Framework in the path.
 - **CUSTOM_ACTION_JAR_PATH**: (Optional) The directory in which the *actions.jar* file is saved. To validate the *actions.jar* file from your current Data Exchange Framework installation, leave this value empty. This value is only required if you want to validate an *actions.jar* file that is not part of your current Data Exchange Framework installation.
5. Save your changes.
6. Navigate to the **validateCustomActionInCU > validateCustomActionViaConfig** folder.
7. Open a command prompt and run the *ValidateCustomActionInCU.bat* file.
The validation tool checks your *actions.jar* file.

- If the file is not compatible with the Data Exchange Framework version to which you are upgrading, an error message appears. You will not be able to upgrade until you provide a valid *actions.jar* file. For example:

Validation Custom Action in Nexidia Data Exchange Framework Multiple Servers
Cumulative Update ...

Success: Administrative permissions confirmed.

Item 1 is: DOMAIN_NAME=<domain>,INSTALL_DIRECTORY=<dir>,CUSTOM_ACTION_JAR_PATH=

Server Type= Service Name=NDEFW2 Version=5.8.217 Data Directory=<dir>

Custom Action Path: \\<domain>\<dir>\data\patches\actions.jar

error level is: 1

there is an error in validation custom action for DOMAIN_NAME=<domain>,INSTALL_DIRECTORY=<dir>,CUSTOM_ACTION_JAR_PATH=.

Failure: error validation "\\<domain>\<dir>\data\patches\actions.jar"

error occurred in Custom action validation process

- If the file is compatible with the Data Exchange Framework version to which you are upgrading, a success message appears. For example:

Validation Custom Action in Nexidia Data Exchange Framework Multiple Servers
Cumulative Update ...

Success: Administrative permissions confirmed.

Item 1 is: DOMAIN_NAME=<domain>,INSTALL_DIRECTORY=<dir>,CUSTOM_ACTION_JAR_PATH=

Server Type= Service Name=NDEFW2 Version=5.8.217 Data Directory=<dir>

Custom Action Path: \\<domain>\<dir>\data\patches\actions.jar

Custom action validation is completed for Item 1: DOMAIN_

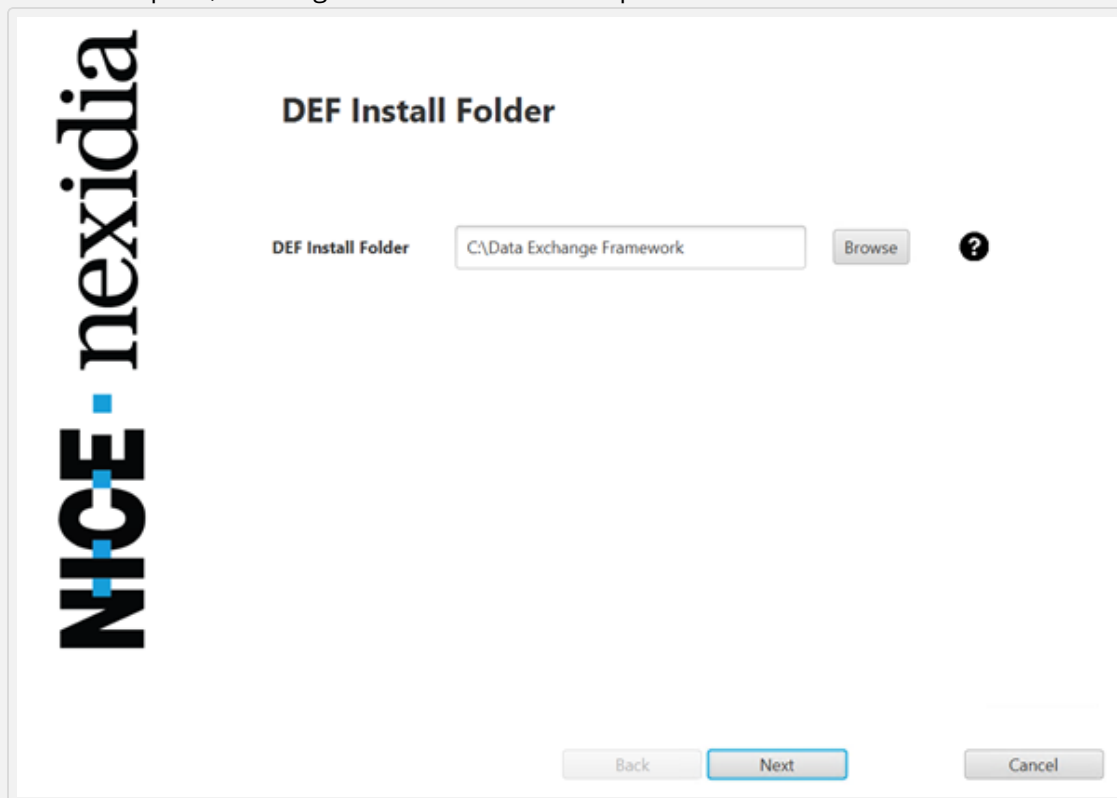
NAME=<domain>,INSTALL_DIRECTORY=<dir>,CUSTOM_ACTION_JAR_PATH=

Custom action validation is completed for all servers

→ To validate custom actions from a user interface:

1. Navigate to the *CumulativeUpdateTool-5.8.zip* file and extract it to a local folder.
2. Navigate to the **validateCustomActionInCU** folder.
3. Open a command prompt and run the *ValidateCustomActionForCumulativeUpdate.bat* file. You are prompted to specify if you want to use the embedded JRE.
4. Enter "y" or "n". If you don't want to use the embedded JRE, then you must have JAVA_HOME in your environment. If you have not set up your JAVA_HOME, you are asked to enter the path to the JRE.

A window opens, showing the DEF Install Folder options.



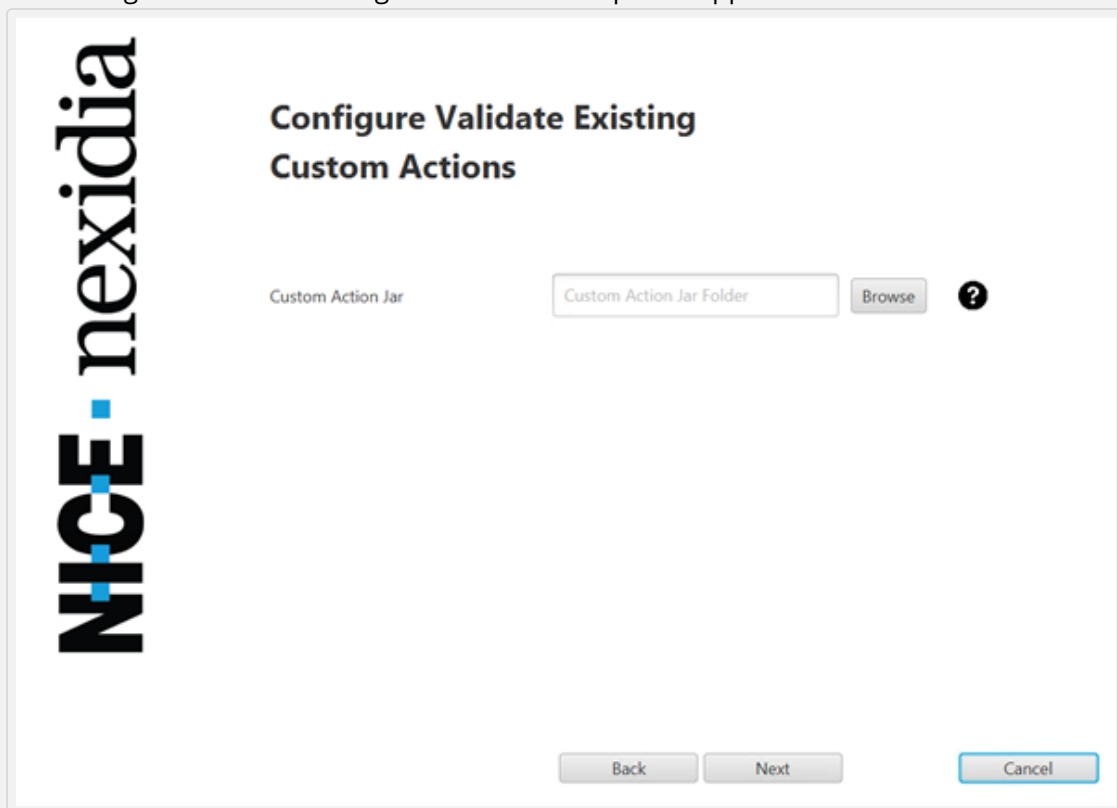
5. Navigate to the Data Exchange Framework installation location and select it. The default location is **C:\Program Files\Nexidia\Data Exchange Framework**.

Important

You must drill down from the top directory of the Installation package folder structure into the sub-directory where the bin, data, etc, install, and jre files are located, and use that complete path for the validation tool to recognize the location.

6. Click **Next**.

The Configure Validate Existing Custom Actions options appear.



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Configure Validate Existing Custom Actions

Custom Action Jar

Custom Action Jar Folder

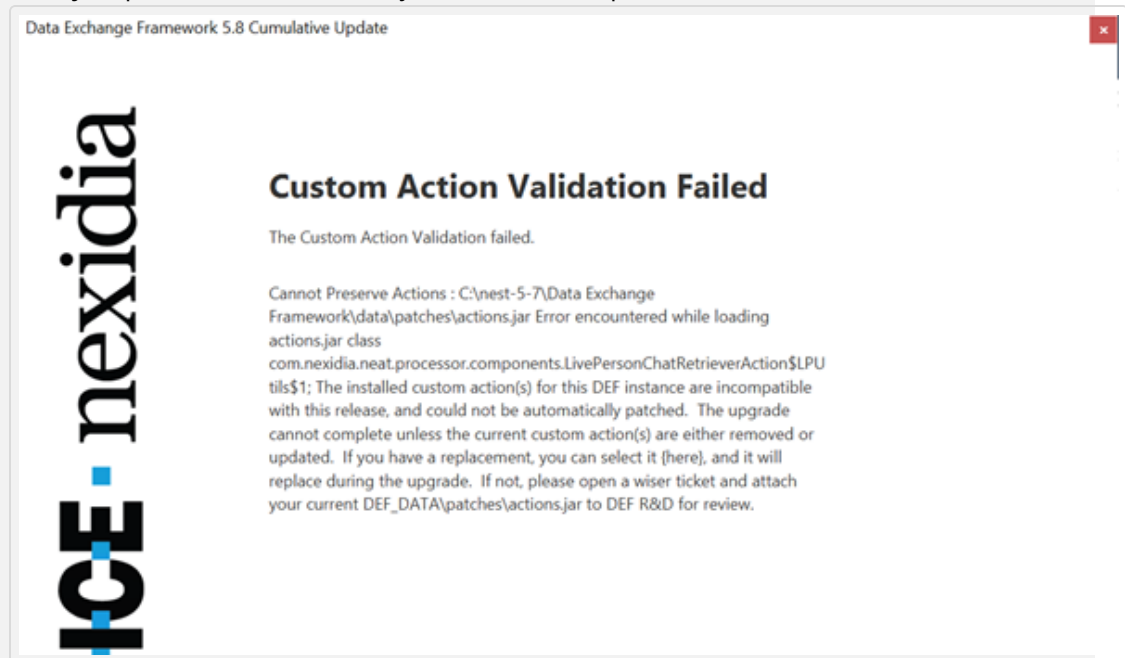
Browse ?

Back Next Cancel

7. Do one of the following:

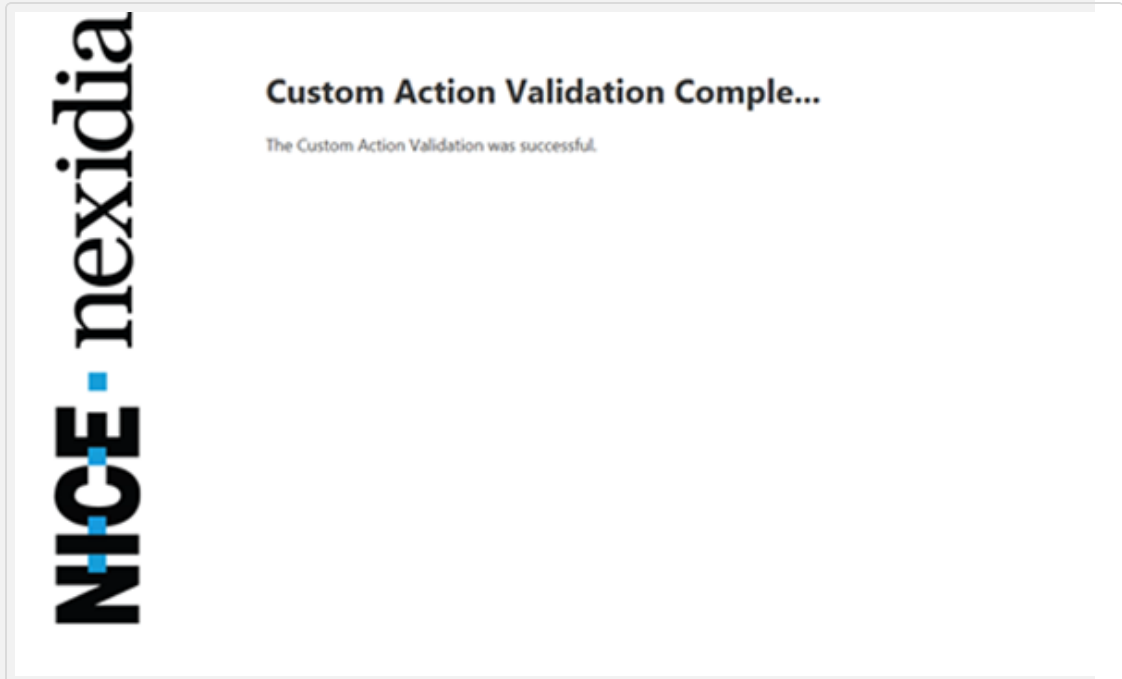
- To validate the *actions.jar* file from your current Data Exchange Framework installation, leave this value empty.

- To validate an *actions.jar* file that is not part of your current Data Exchange Framework installation, click **Browse**, then browse to and select the *actions.jar* file.
8. Click **Next**.
- The validation tool checks your *actions.jar* file.
- If the file is not compatible with the Data Exchange Framework version to which you are upgrading, an error message appears. You will not be able to upgrade until you provide a valid *actions.jar* file. For example:



- If the file is compatible with the Data Exchange Framework version to which you

are upgrading, a success message appears. For example:

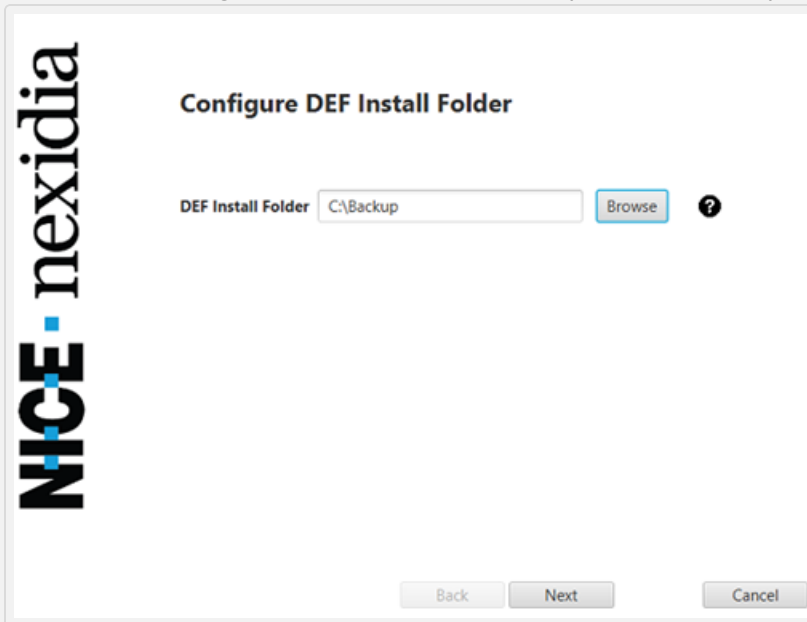


UPGRADING AN INDIVIDUAL SERVER

→ To upgrade an individual server:

1. Stop all running processes and allow them to wind down.
2. Shut down the Nexidia Data Exchange Framework service.
3. In the Task Manager, make sure that the Java processes have terminated.
4. Navigate to the *CumulativeUpdateTool-X.X.zip* file and extract it to a local folder.
5. Open a command prompt and run the *CumulativeUpdate.bat* file.
You are prompted to specify if you want to use the embedded JRE.
6. Enter **"y"** or **"n"**. If you don't want to use the embedded JRE, then you must have JAVA_HOME in your environment. If you have not set up your JAVA_HOME, you are asked to enter the path to the JRE.

The Data Exchange Framework Cumulative Update window opens.

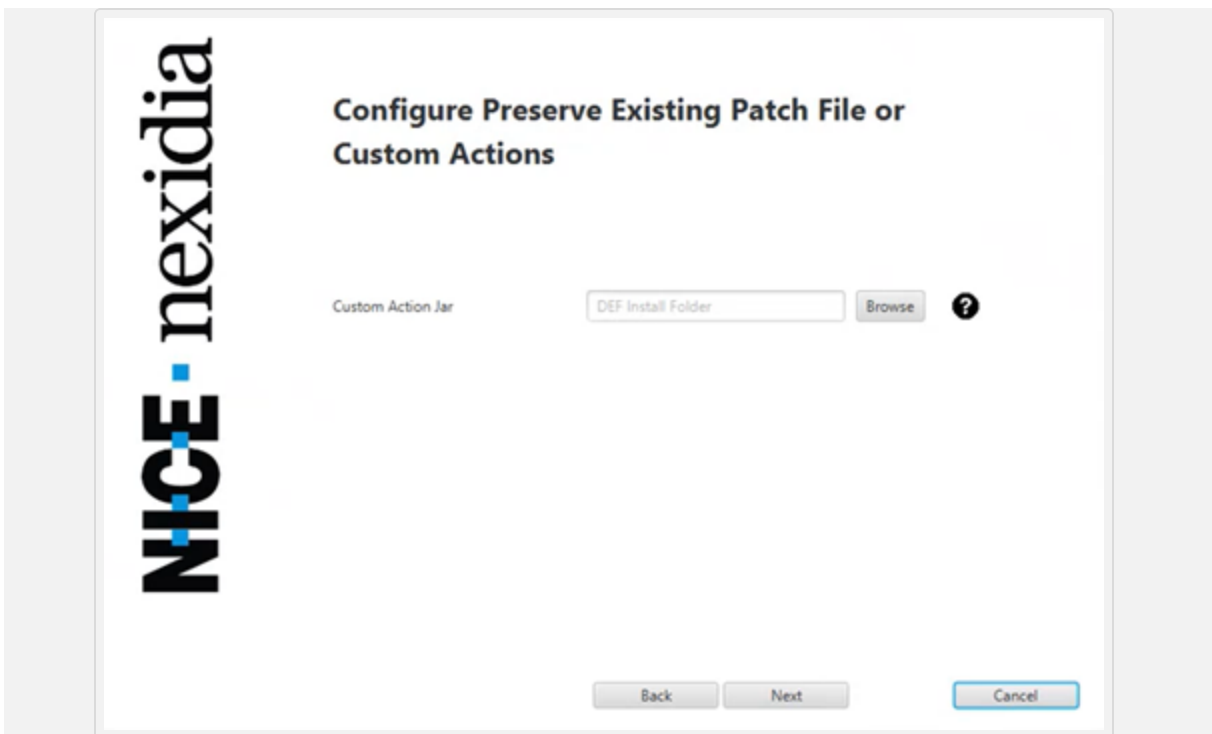


7. Navigate to the Data Exchange Framework installation location and select it. The default location is **C:\Program Files\Nexidia\Data Exchange Framework**.

Important

You must drill down from the top directory of the Installation package folder structure into the sub-directory where the bin, data, etc, install, and jre files are located, and use that complete path for the Cumulative Updater to recognize the location.

8. Click **Next**.
The Configure Preserve Existing Patch File or Custom Actions options appear.

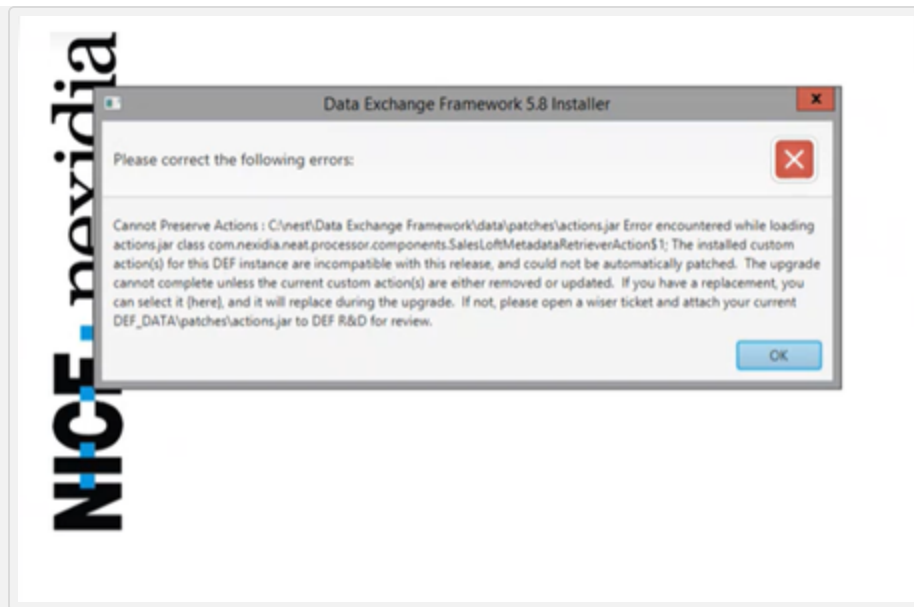


9. Do one of the following:

- To keep the *actions.jar* file from your current Data Exchange Framework installation, leave this value empty.
- To replace the *actions.jar* file from your current Data Exchange Framework installation with a new file, click **Browse**, then browse to and select the new *actions.jar* file.

10. Click **Next**.

- If the *actions.jar* file is not compatible with the Data Exchange Framework version to which you are upgrading, an error message appears. You will not be able to upgrade until you provide a valid *actions.jar* file.



- If the *actions.jar* file is compatible with the Data Exchange Framework version to which you are upgrading, Authentication Strategy options appear.

11. Click **Next**.

The Configure Disable TLS 1.0/1.1 options appear.



12. Select the checkbox to disable TLS 1.0/1.1.
If you select this option, the Cumulative Updater appends the values: SSLv2Hello, SSLv3, TLSv1, TLSv1.1 for the tag: `jdk.tls.disabledAlgorithms` in the `jre/lib/security/java.security` file.
13. Click **Next**.
The Confirm Cumulative Update options appear.



Confirm Cumulative Update

Confirm your specified values and then press 'Next' to start the installation.

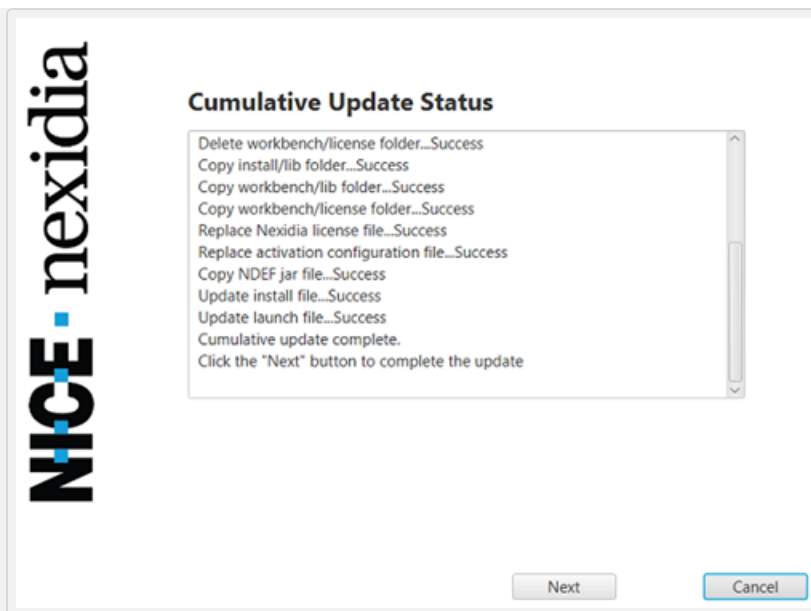
DEF Install Folder	C:\nest-5-7\Data Exchange Framework
TLS Disabled	false
Preserve Custom Actions	true

Back Next Cancel

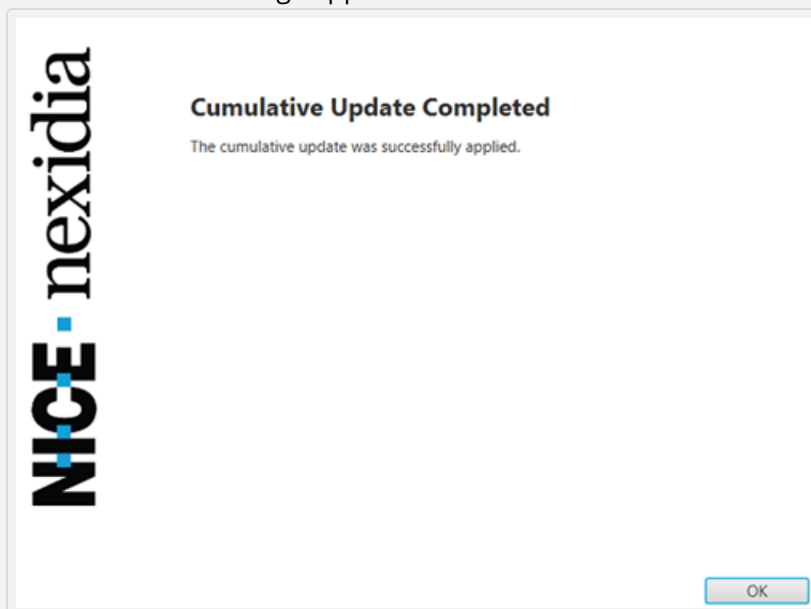
Important

If your custom actions can be replaced with core actions, **Preserve Custom Actions** will be set as **false**, because the Cumulative Updater will delete the custom actions. **Preserve Custom Actions** will also be set as **false** if you are replacing the *actions.jar* file with a new file.

- Click **Next**.
The Cumulative Update Status appears.



15. When the update is complete, click **Next**.
A confirmation message appears.



Important

As part of the upgrade to Data Exchange Framework 5.8 or later, the Cumulative Updater replaces the *log4j.properties* file in the **/etc** folder with the updated *log4j2.properties* file.

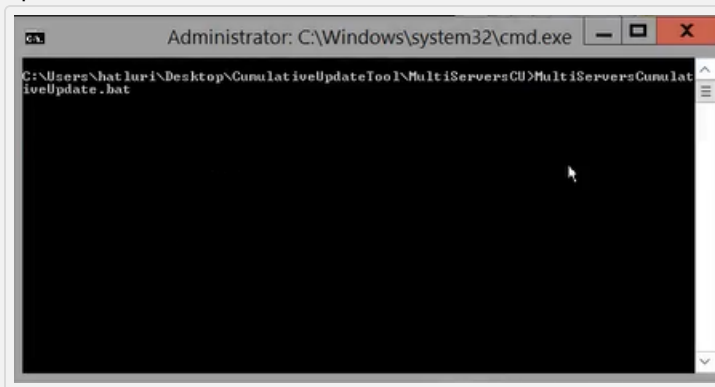
16. Click **OK**, then start the Data Exchange Framework service and re-schedule the processes.

UPGRADING MULTIPLE SERVERS

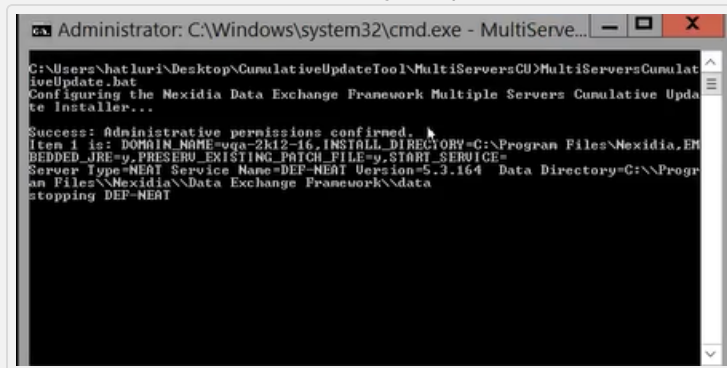
→ To upgrade multiple servers:

1. Navigate to the *CumulativeUpdateTool-X.X.zip* file and extract it to a local folder.
2. Navigate to the **MultiServersCU > config** folder.
3. Open the *batchCUTemplateproperties* file. This file is empty when you initially extract it.
4. Enter values to specify where each instance of DEF to be updated is located. You can add as many lines as you need based on how many servers you have. You must include these values:
 - **DOMAIN_NAME**: The server name.
 - **INSTALL_DIRECTORY**: The directory where the Data Exchange Framework is installed. For Example c:\Program Files\Nexidia. Do not include Data Exchange Framework in the path.
 - **EMBEDDED_JRE**: Enter **y** if the deployment uses the embedded JRE or **n** if it does not.
 - **START_SERVICE**: If you want to start the service after the update, leave **blank** or enter **y**. If you do not want to start the service after the update, enter **n**.
 - **CLUSTER_ENVIRONMENT**: Set to **y** if it is cluster. This is responsible for stopping/service from the cluster failover manager.
 - **TLS_DISABLED**: Set to **y** to append the value, or **n**, to not append the value.

- **CUSTOM_ACTION_JAR_PATH:** (Optional) The directory in which a new custom actions (*actions.jar*) file is saved. This value is only required if you want to replace the *actions.jar* file from your current Data Exchange Framework installation with a new file.
5. Open a command prompt and run the *MultiServersCumulativeUpdate.bat* to start the updates.



6. Each step in the process is logged for each server.
- The DEF service is stopped.
 - If the *actions.jar* file is not compatible with the Data Exchange Framework version to which you are upgrading, an error message appears. You will not be able to upgrade until you provide a valid *actions.jar* file.
 - The required files are updated.
 - The DEF service is restarted (if you specified for it to restart).



- When all of the servers are updated, the Cumulative Updater stops and the log is updated to indicate that the update is completed for all servers.

```

CU
replace \\vqa-DEF-04\C$\DEF\Data Exchange Framework\data\workbench\lib with CU
replace \\vqa-DEF-04\C$\DEF\Data Exchange Framework\data\workbench\license with
CU
replace \\vqa-DEF-04\C$\DEF\Data Exchange Framework\data\workbench\license\Nexid
ialicense.lic with the existing one
replace \\vqa-DEF-04\C$\DEF\Data Exchange Framework\data\workbench\license\activ
ationConfiguration.xml with the existing one
replace \\vqa-DEF-04\C$\DEF\Data Exchange Framework\lib\ndef-5.3.jar with CU
update version number to in 5.3.166 in \\vqa-DEF-04\C$\DEF\Data Exchange Framewo
rk\etc\ndef.properties
1 file(s) moved.
update \\vqa-DEF-04\C$\DEF\Data Exchange Framework\bin\restartNow.bat
update \\vqa-DEF-04\C$\DEF\Data Exchange Framework\bin\uninstallService.bat
update \\vqa-DEF-04\C$\DEF\Data Exchange Framework\bin\launch.bat
delete all the copy folders
starting CU-DEF
Update is completed for Item 4 : DOMAIN_NAME=vqa-DEF-04,INSTALL_DIRECTORY=C:\DEF
,EMBEDDED_JRE=y,PRESERV_EXISTING_PATCH_FILE=y,START_SERVICE=
update is completed for all servers.
Press enter to exit

C:\Users\hatluri\Desktop\CumulativeUpdateTool\MultiServersCU>

```

Important

As part of the upgrade to Data Exchange Framework 5.8 or later, the Cumulative Updater replaces the *log4j.properties* file in the **/etc** folder with the updated *log4j2.properties* file. In addition, if your custom actions can be replaced with core actions, the Cumulative Updater deletes the custom actions.

- Press **enter** to exit the command prompt.

UPGRADING A CLUSTERED DEF ENVIRONMENT

You can use the Cumulative Updater to upgrade DEF 5.0 or greater installed on a failover (Microsoft Cluster) environment. The process includes the following steps:

- [Stop the DEF Processes](#) on the facing page: Stop the DEF Administrator and Worker services on each node in the cluster.
- [Run the Cumulative Updater to Upgrade a Clustered DEF Environment](#) on page 98: Run the Cumulative updater on each node in the cluster, for each service.
- [Restart the DEF Processes](#) on page 103: Restart the DEF Administrator and Worker services on each node in the cluster.

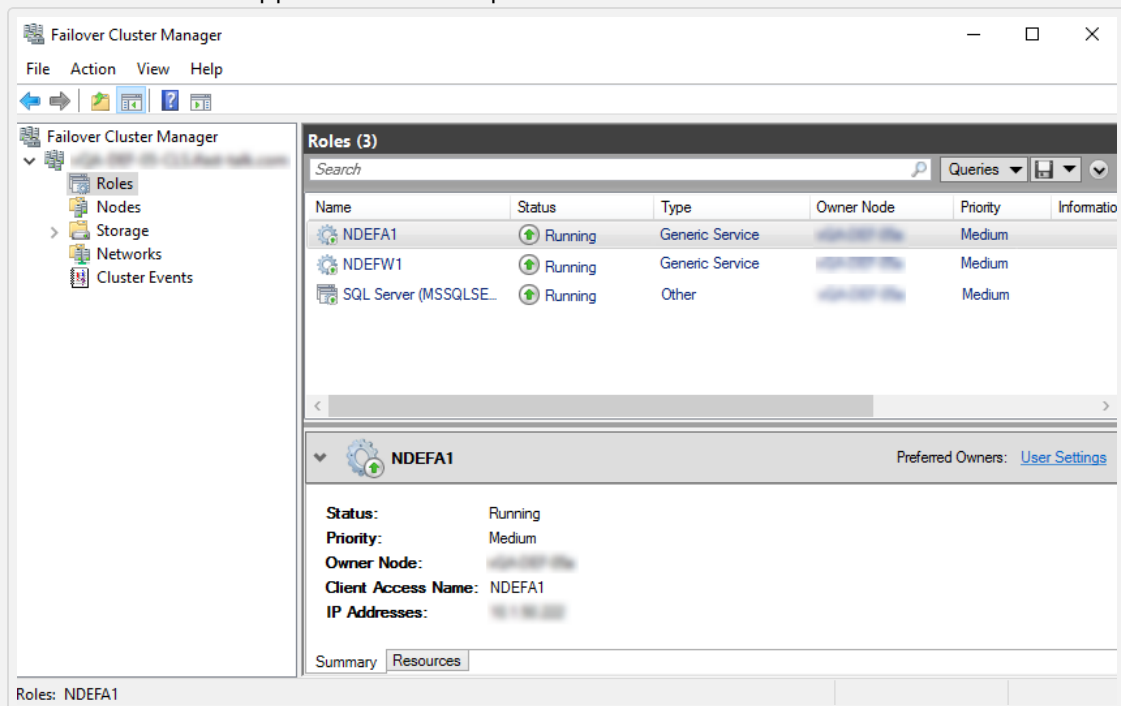
4. [Verify the DEF Version](#) on page 104: Open the DEF user interface and verify that the updated version appears.

Stop the DEF Processes

Before you run the Cumulative Updater to upgrade the servers in a clustered DEF environment, you must stop all DEF processes by stopping the Administrator and Worker roles in the Failover Cluster Manager. If you do not stop either of these roles, an error message will appear when you try perform the upgrade.

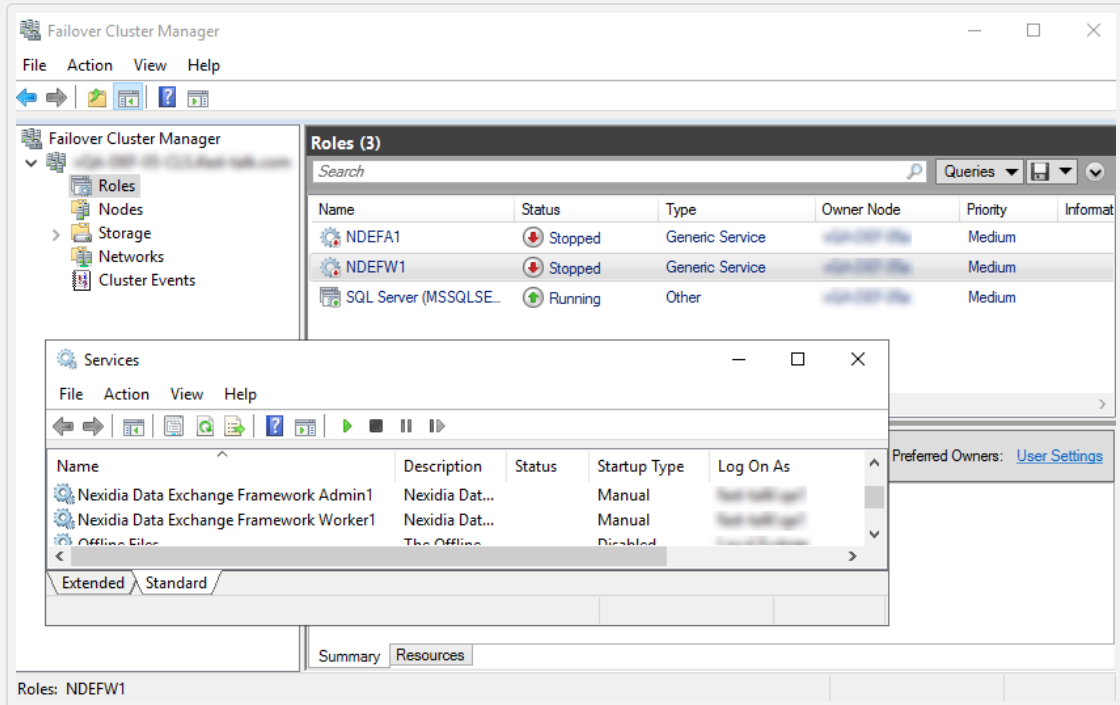
→ To stop the DEF processes:

1. In the Failover Cluster Manager, browse to **<Cluster> > Roles**.
The available roles appear in the Roles pane.



2. Stop the Administrator role by right-clicking on the role and selecting **Stop Role**.
The status of the role changes to **Stopped**.

3. Stop the Worker role by right-clicking on the role and selecting **Stop Role**.
The status of the role changes to **Stopped**.
4. Ensure that the Administrator and Worker services are stopped on each cluster node.



Run the Cumulative Updater to Upgrade a Clustered DEF Environment

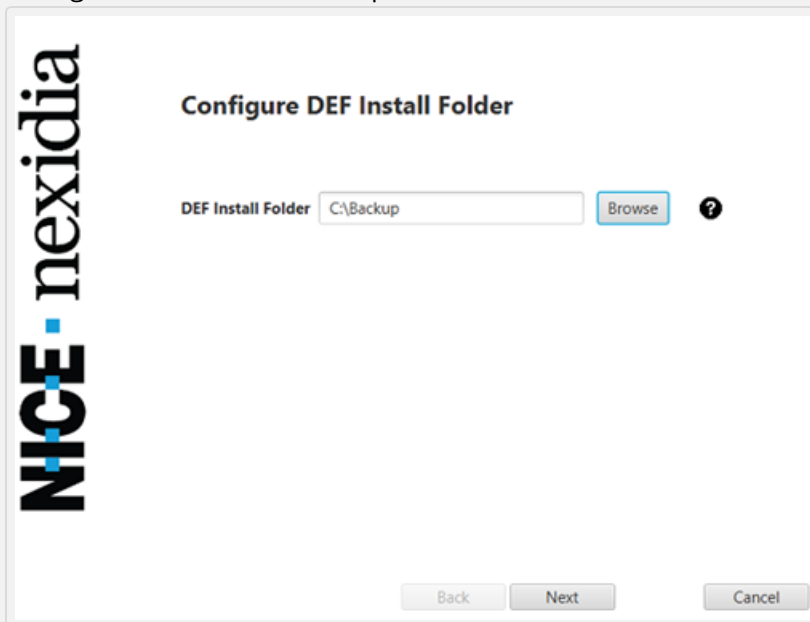
To upgrade a clustered DEF environment, you must run the Cumulative Updater on each node, for each service.

Warning

Do not upgrade the same services on both nodes in parallel. During the upgrade, the Cumulative Updater replaces shared files that are common for both cluster nodes, and upgrading the services on both nodes in parallel can cause an exception.

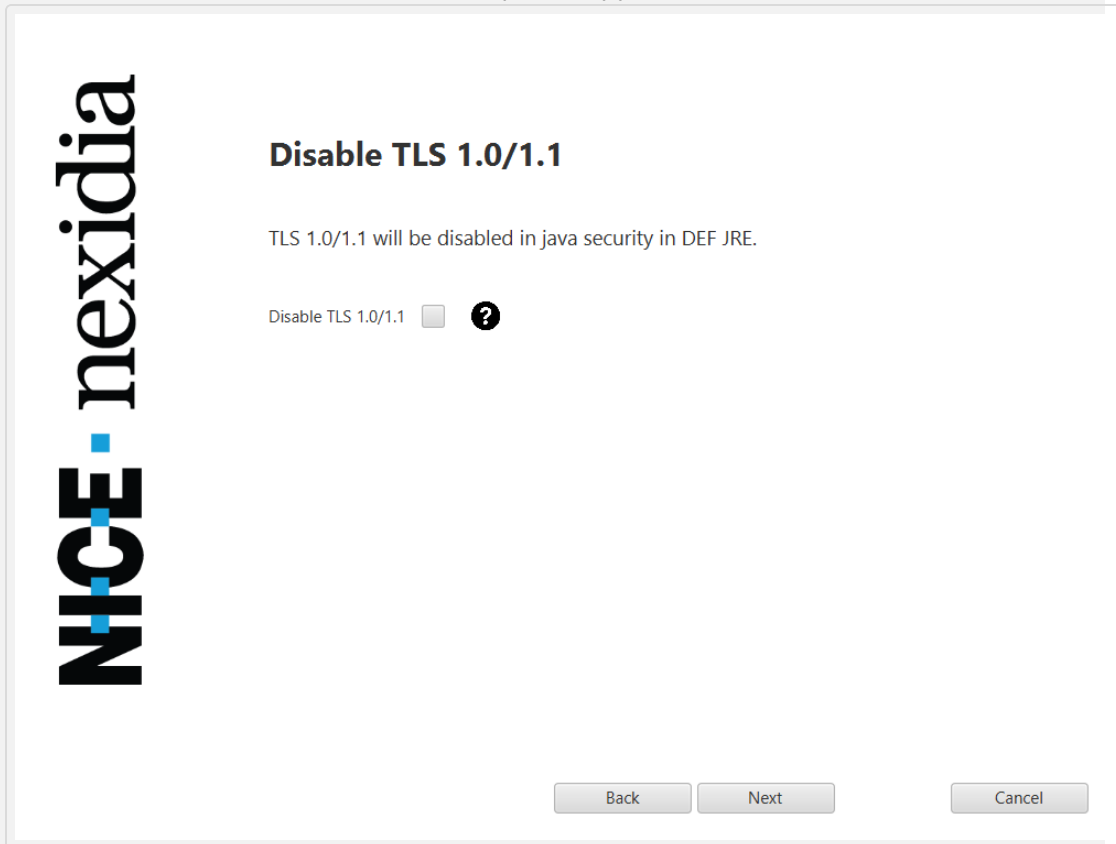
→ To run the Cumulative Updater to upgrade a clustered DEF environment:

1. Navigate to the *CumulativeUpdateTool-X.X.zip* file and extract it to a local folder on each machine. It is recommended to keep the directory structure as short as possible.
2. Browse to **CumulativeUpdateTool-X.X > CumulativeUpdateTool**, and run the *CumulativeUpdate.bat* file as an administrator.
You are prompted to specify if you want to use the embedded JRE.
3. Enter “y” or “n”. If you don't want to use the embedded JRE, then you must have JAVA_HOME in your environment. If you have not set up your JAVA_HOME path, you are asked to enter the path to the JRE.
The Data Exchange Framework Cumulative Update window opens, showing the Configure DEF Install Folder options.

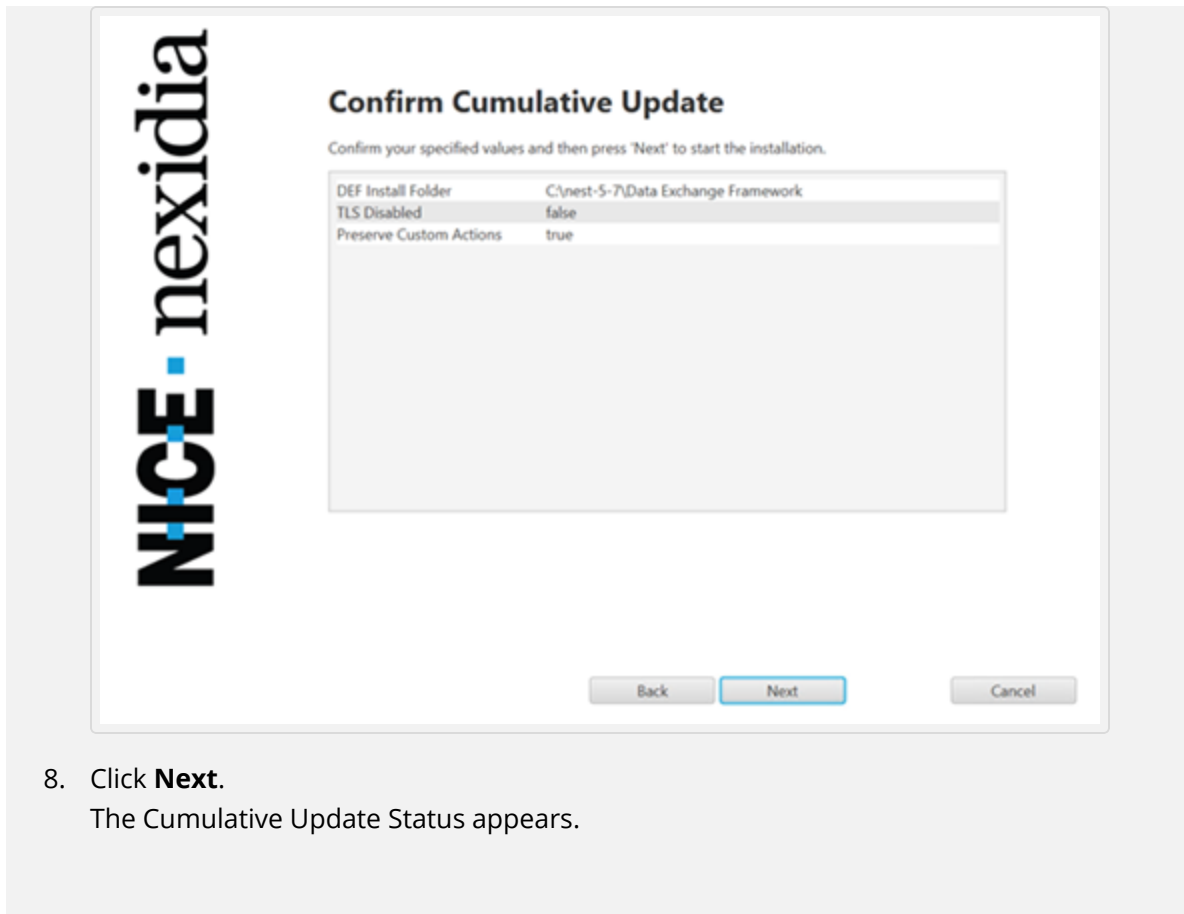


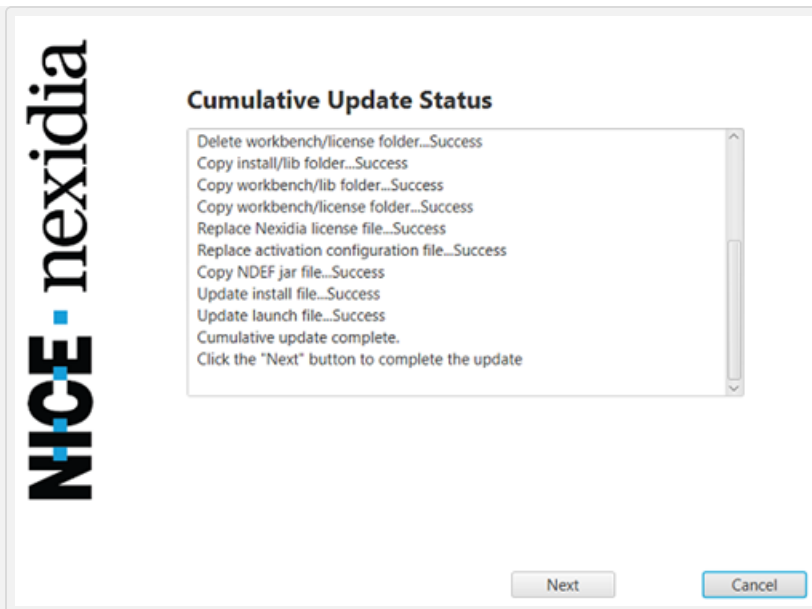
4. Navigate to the Data Exchange Framework installation directory and select it. You must drill down from the top directory of the Installation package folder structure into the sub-directory where the bin, data, etc, install, and jre files are located, and use that complete path for the Cumulative Updater to recognize the location.
5. Click **Next**.

If the embedded JRE folder doesn't already have TLS disabled for SSLv2Hello, SSLv3, TLSv1, TLSv1.1, the Disable TLS 1.0/1.1 options appear.

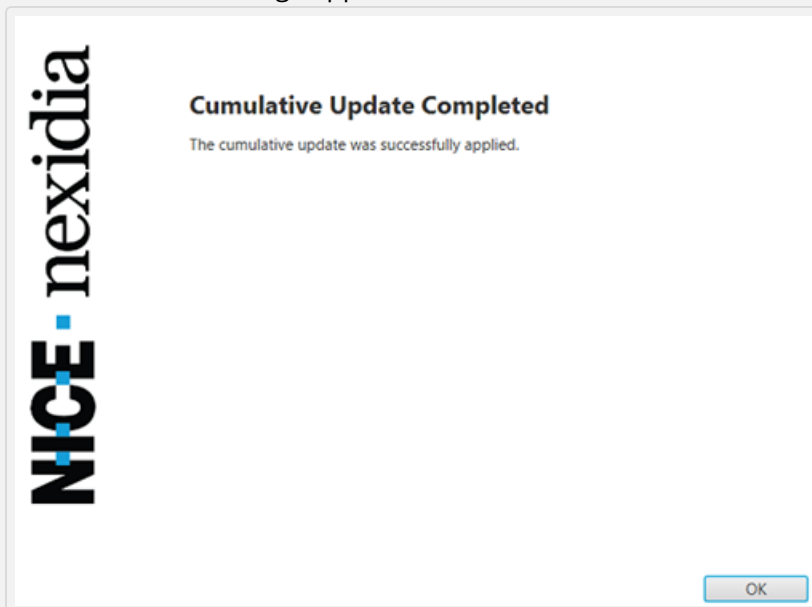


6. Select the checkbox to disable TLS 1.0/1.1.
The Cumulative Updater appends the values: SSLv2Hello, SSLv3, TLSv1, TLSv1.1 for the tag: `jdk.tls.disabledAlgorithms` in the `jre/lib/security/java.security` file.
7. Click **Next**.
Confirm Cumulative Update options appear.





9. When the update is complete, click **Next**.
A confirmation message appears.



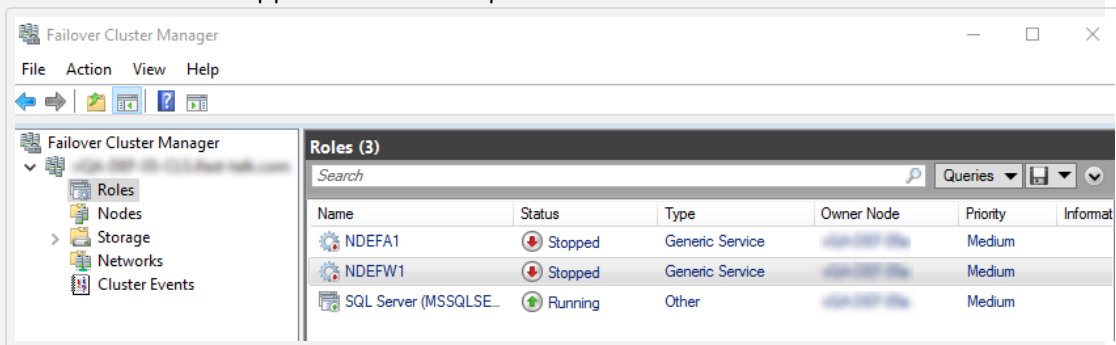
10. Click **OK**.

Restart the DEF Processes

After you run the Cumulative Updater to upgrade the servers in a clustered DEF environment, you must restart all DEF processes by starting the Administrator and Worker roles in the Failover Cluster Manager. It is recommended to check the Windows Path variable before restarting the services.

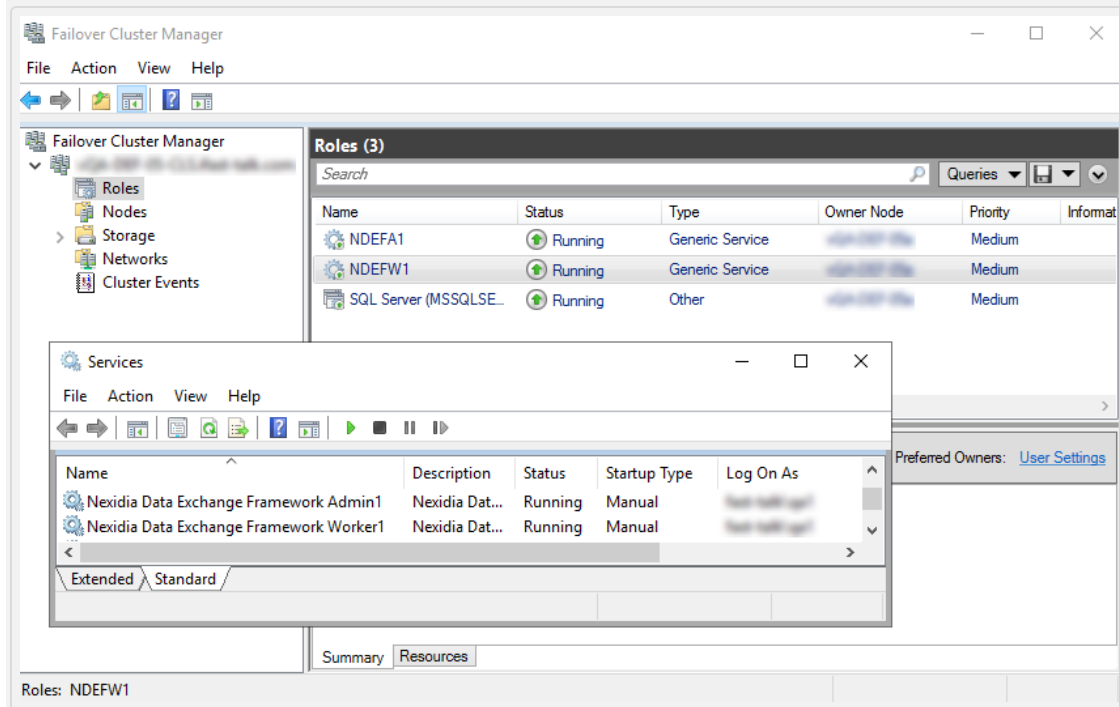
→ To start the DEF processes:

1. In the Failover Cluster Manager, browse to **<Cluster> > Roles**.
The available roles appear in the Roles pane.



2. Start the Administrator role by right-clicking on the role and selecting **Start Role**.
The status of the role changes to **Running**.
3. Start the Worker role by right-clicking on the role and selecting **Start Role**.
The status of the role changes to **Running**.
4. Ensure that the Administrator and Worker services are up and running on the primary

node.



Verify the DEF Version

After completing the upgrade and restarting the DEF services, open the DEF user interface and verify that the updated version appears.

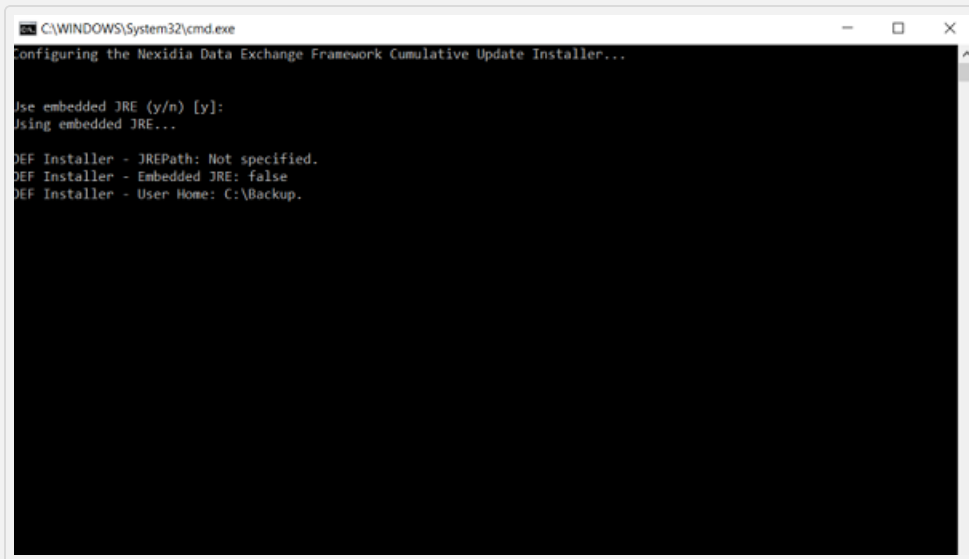
➔ To verify the DEF version:

1. In your web browser, clear the browser cache.
2. Go to **http://<cluster group name>:8080/**.
The DEF sign in page appears.
3. Check that the version number that appears below the Sign In button matches the version to which you upgraded.

Uninstalling the Application

→ To uninstall the application:

1. Navigate to the **bin** folder in the software file location.
2. Right-click on the **uninstallService.bat** file and select **Run as administrator**. The user interface opens and you are prompted to specify if you want to use the embedded JRE or not.

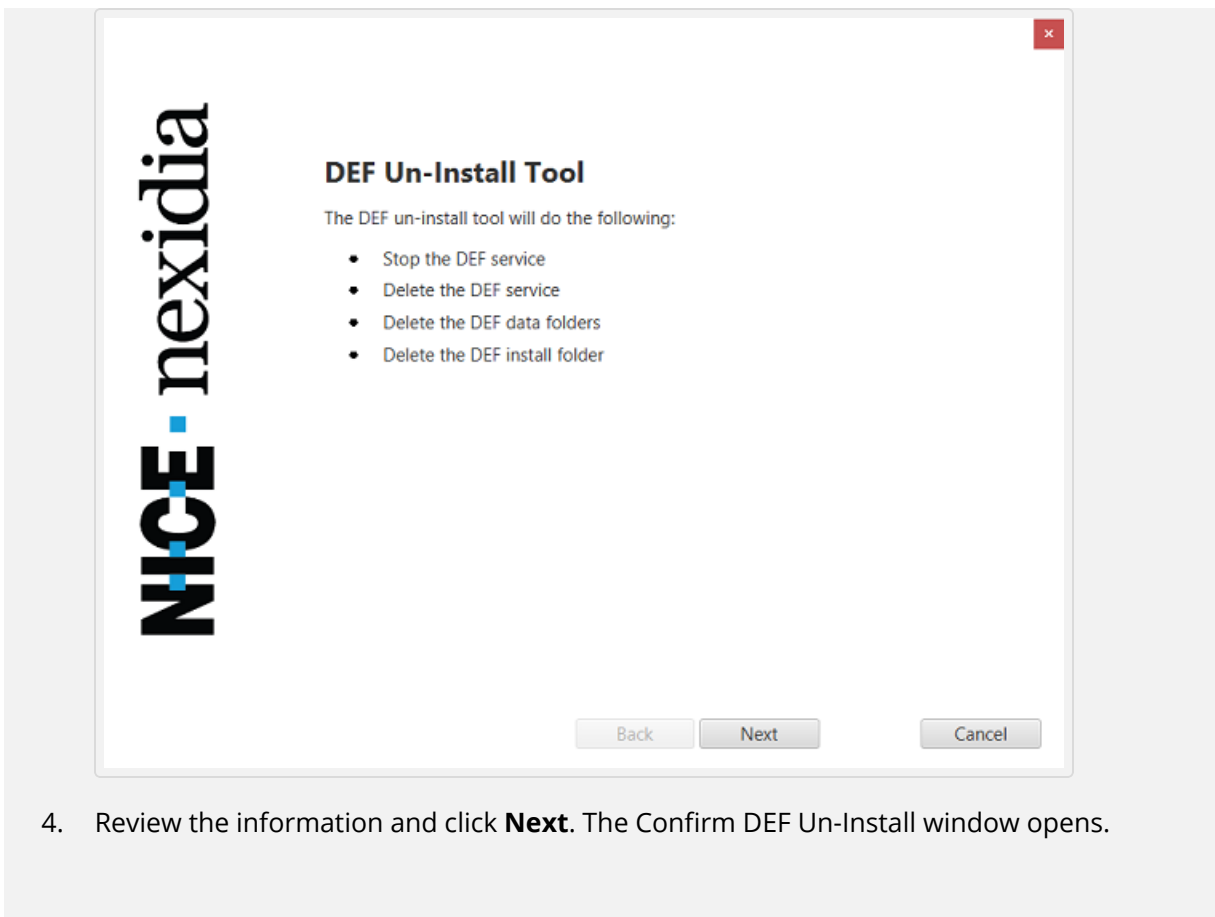


```
C:\WINDOWS\System32\cmd.exe
Configuring the Nexidia Data Exchange Framework Cumulative Update Installer...

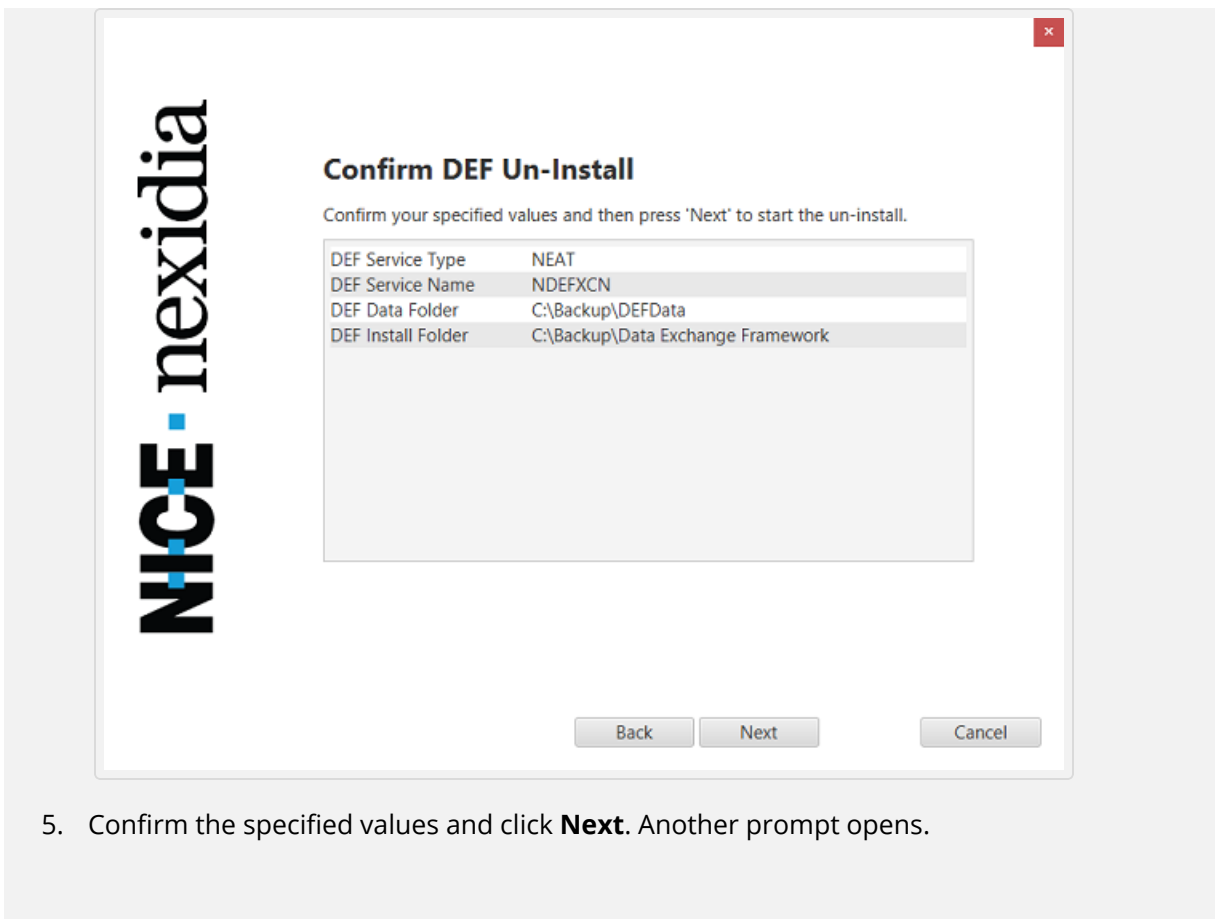
Use embedded JRE (y/n) [y]:
Using embedded JRE...

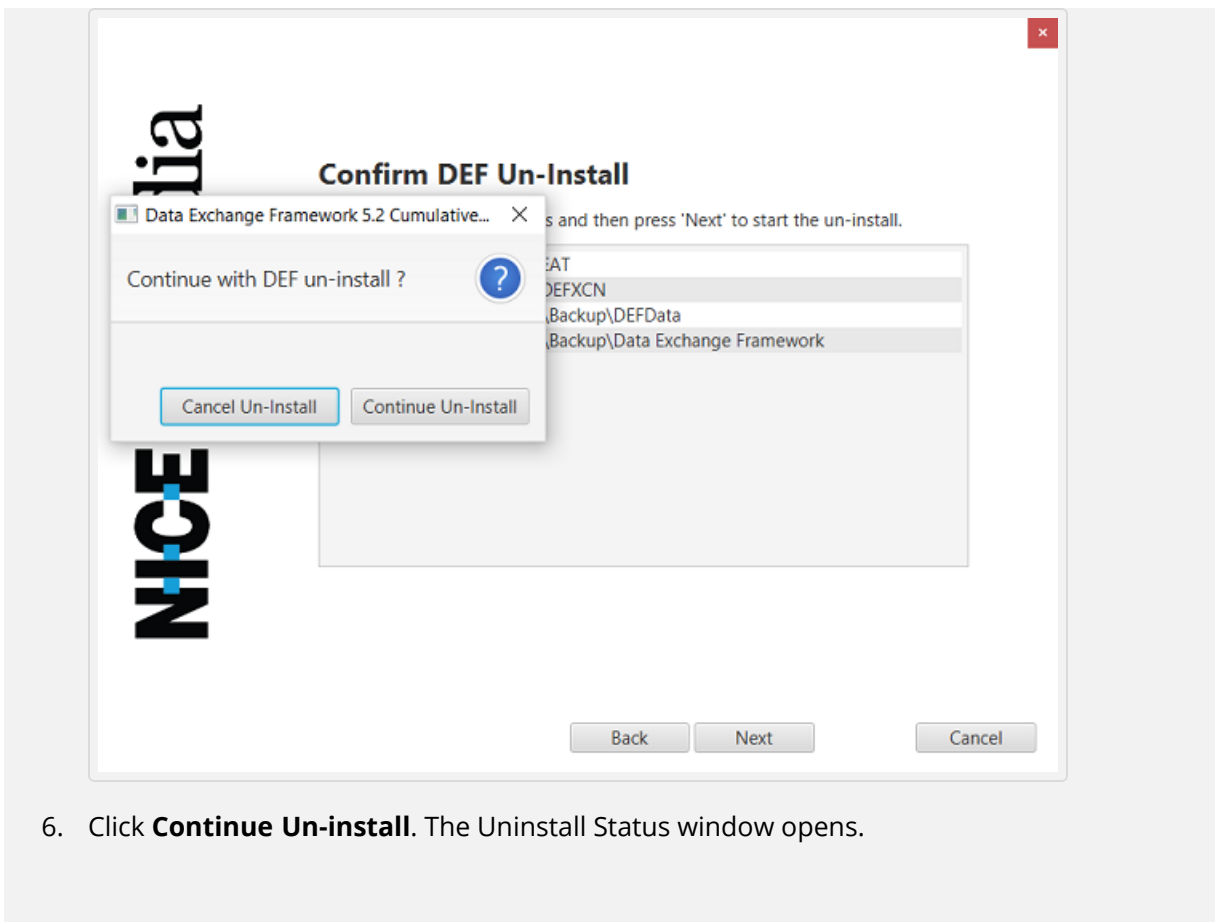
DEF Installer - JREPath: Not specified.
DEF Installer - Embedded JRE: false
DEF Installer - User Home: C:\Backup.
```

3. Enter **"y"** or **"n"**. If you don't want to use the embedded JRE, you must have JAVA_HOME in your environment. If you have not set up your JAVA_HOME, you are asked to enter the path to the JRE. After you make the emedded JRE selection, the DEF Un-Install Tool window opens.

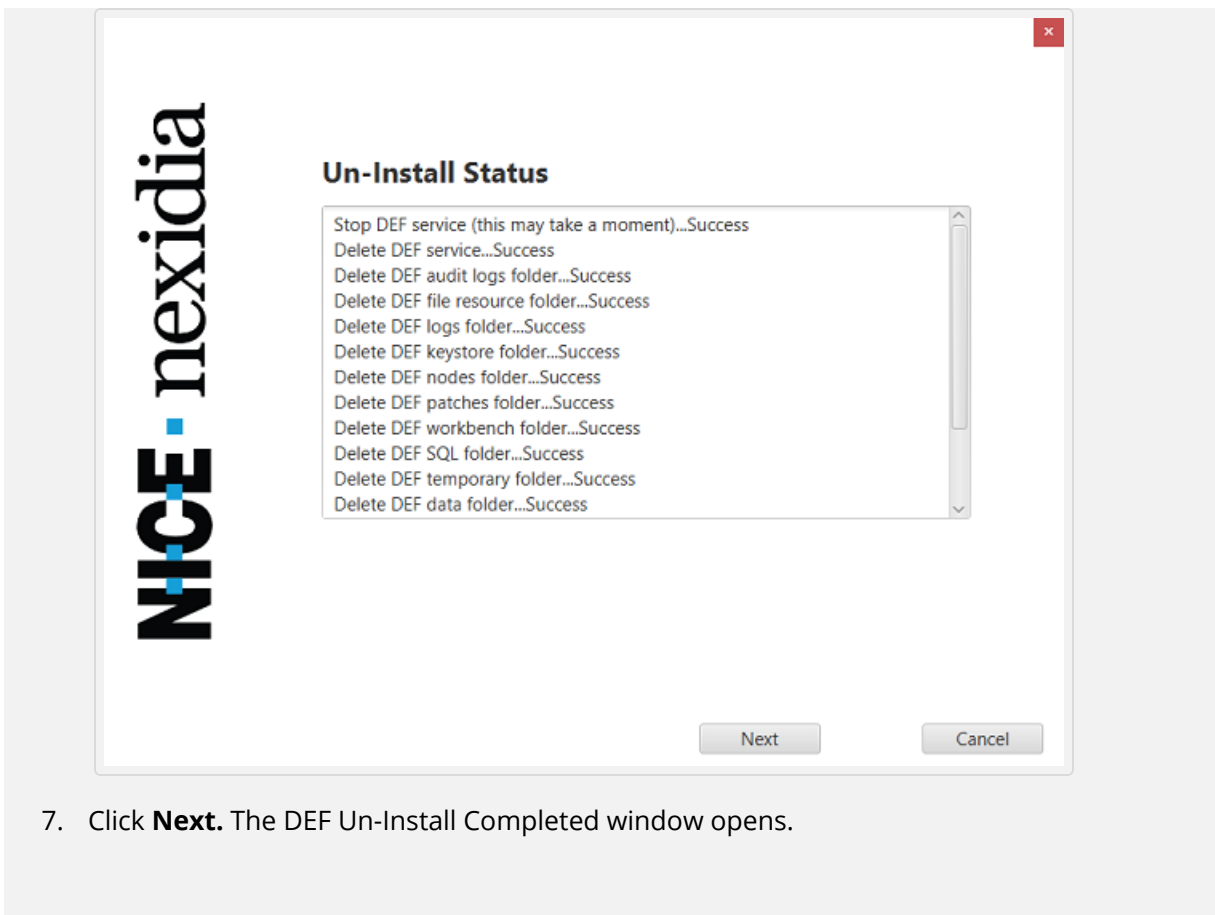


4. Review the information and click **Next**. The Confirm DEF Un-Install window opens.

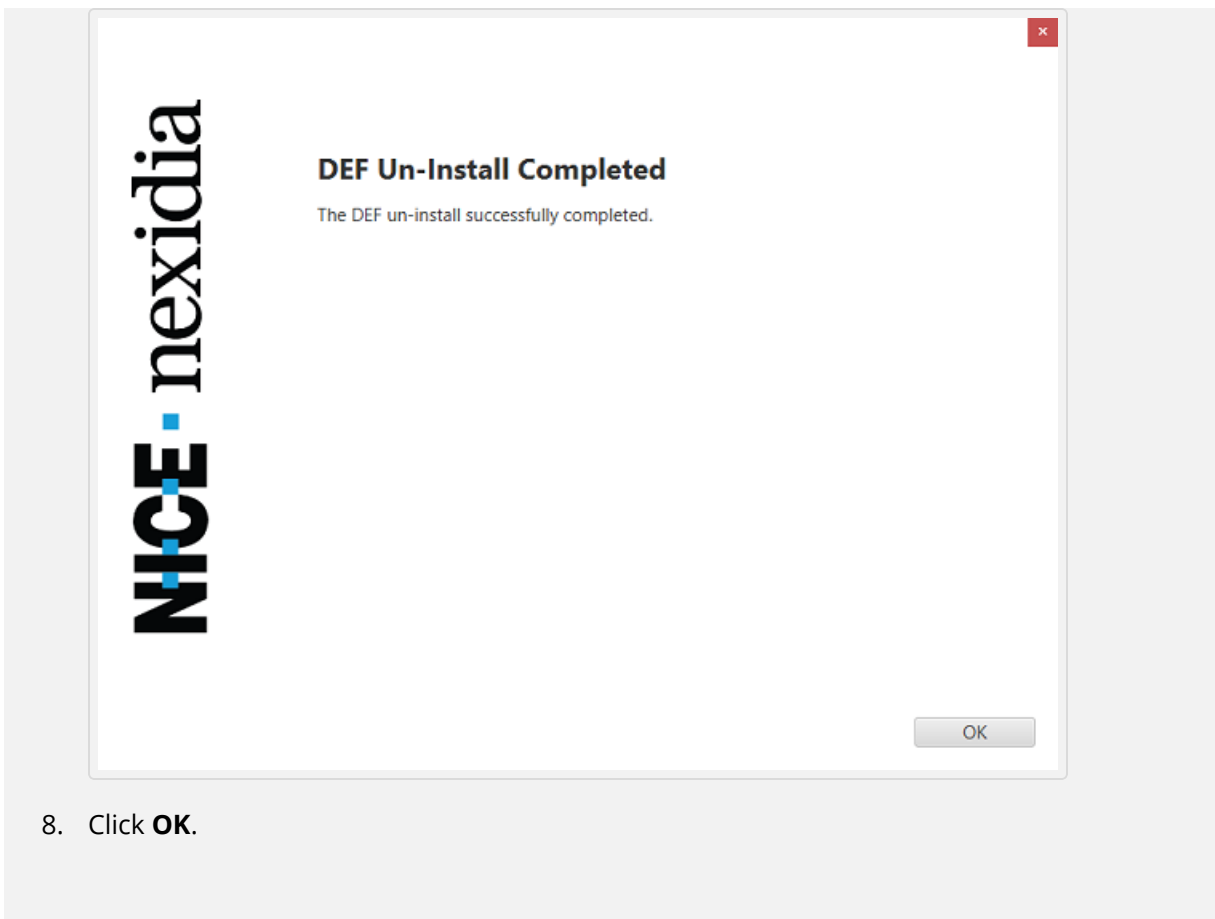




6. Click **Continue Un-install**. The Uninstall Status window opens.



7. Click **Next**. The DEF Un-Install Completed window opens.



8. Click **OK**.

Troubleshooting

This section provides troubleshooting information for various issues related to installing and using the Data Exchange Framework. The application log files are located here: **INSTALL_DATA_nx_def.*.log**.

COMMON INSTALLATION ISSUES

Startup logs have some expected errors regarding foreign key creation by the embedded EclipseLink ORM library, and these can be disregarded. Other exceptions noted as WARN, ERROR, and FATAL in the logs are important for troubleshooting. Java exceptions are presented in stack traces that typically end in a final "Caused by: ..." message. You should capture the full stack trace for review to determine the root cause of an issue.

ISSUE	SOLUTION
When you open a file that has been written by the DEF, the file seems to be corrupted, in binary format, or encrypted.	This is most likely a Vormetrics issue because the currently logged in user is not in the Vormetrics security profile for the specific machine. In this case, when you try to open a file, it seems to be corrupted but it is actually encrypted. Contact IT to add the current user to the Vormetrics security profile.
An error occurs when installing the Microsoft Visual C++ 2015 Redistributable. The error says that a later version of the Microsoft Visual C++ Redistributable is already installed and Microsoft Visual C++ 2015 Redistributable cannot be installed.	1. Uninstall the Microsoft Visual C++ 2017 Redistributable. 2. Install the Microsoft Visual C++ 2015 Redistributable.
After installation, the DEF service won't not start.	Verify the DEF service user account has full access to the {DEF install directory} and {DEF data directory}.

ISSUE	SOLUTION
This message displays during NEST server startup: <pre>ERROR com.nexidia.neat.init.task.NEST.ServerPayloadHandlerTask.call:136 - Error in run() java.lang.NullPointerException at com.nexidia.neat.init.task.NEST.ServerPayloadHandlerTask.call (ServerPayloadHandlerTask.java:##)</pre>	This exception occurs due to NEAT/NEST synchronization issues during NEST's first startup. This will recover with no action necessary after a couple of heartbeats.

DIAGNOSTIC TOOLS

The following tools are provided for troubleshooting in the DEF_DATA\tools folder:

- JDK.zip - Contains the latest Azul JDK matching the Data Exchange Framework internal Java version. You can install the Data Exchange Framework pointing to the JRE folder inside the JDK.
- From the latest Workbench matching the Data Exchange Framework internal version:
 - MediaConvert.zip - Extract and run the MediaConvert.exe using the following command : -tc -files "E:\temp\DtagsMedia\6758763963819819010_6758763963819819011.nmf" -outputdir "E:\temp".
 - Nexidia.FileInfoTool.zip - Extract and run the FileInfotool.exe to see the details of the media files.